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Steering Turret Coral

Preliminary Design Justification File
(Deliverable DRL 15)

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Change Log

Revision	Section	Changes
1.0	First Issue	
2.0	§ 3.3	Pixel/distance camera parameters Included the BIT timeout and frequency Deleted the External PC Described the Manual mode Updated transition description (figure 7) AI #2 ALMA-STC-MoM-005
	§ 3.4	Updated Turret description

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Acronyms and Definitions

A	
AIT	Assembly & Integration Tests
AIV	Assembly & Integration Verification
C	
CCB	Configuration Control Board
CM	Configuration Management
CoC	Certificato di conformità
COTS	Commercial Off-the-Shelf component
CP	Change Proposal
CP	Check Point
CSCI	Computer Software Configuration Item
CU	Control Unit
D	
DOF	Degree of Freedom
F	
FoV	Field of View
H	
HMI	Human-Machine interface
IDD	Interface Design Description
I	
IRS	Interface Requirements Specification
L	
LoS	Line of Sight
M	
MGR	Matrice di Gestione dei Requisiti
N	
NC	Non Conformità
NR	Nota di Rilascio
NT	Nota Tecnica
O	
OdA	Ordine di Acquisto
OSL	Optically Stimulated Luminescence
P	
PC	Personal Computer
PDF	Portable Document Format
PDQ	Piano di Qualità
PM	Program Manager
PR	Problem Report
Q	
QM	Quality Manager
QMS	Quality Management System
R	
RdA	Richiesta di Acquisto.
RFC	Request for Clarification
S	
SCP	System Change Proposal
SDD	Software Design Description
SIA	Scuola di Ingegneria Aerospaziale
SPR	System Problem Report
SRS	Software Requirements Specification
SSS	System Subsystem Specification
ST	Steering Turret
STD	Software Test Description
STR	Software Test Report
SUM	Software User Manual
SUT	System Under Test
SVD	Software Version Description
T	
TBC	To be Confirmed
TBD	To Be Defined
V	
VDR	Verbale di Riunione
W	
WP	Work Package

Table 1 - Acronyms

COTS	Commercial Off-the-Shelf component - Si riferisce a componenti hardware e software disponibili sul mercato.
FoV	The field of view (FOV) is the angular extent of the observable world that is seen at any given moment. In the case of optical instruments or sensors, it is a solid angle through which a detector is sensitive to electromagnetic radiation. It is further relevant in photography.
LoS	The line of sight, also known as visual axis or sightline (also sight line), is an imaginary line between a viewer/observer/spectator's eye(s) and a subject of interest, or their relative direction. The subject may be any definable object taken note of or to be taken note of by the observer, at any distance more than least distance of distinct vision. In optics, refraction of a ray due to use of lenses can cause distortion

Table 2 - Definitions

1 SCOPE OF THE DOCUMENT

1.1 IDENTIFICATION

This document is the Design Justification File. It describes the main design decision for the Steering Turret Coral project – PHASE A). It is identified as ALMA-STC-DJF-014.

This document refers other documents that are part of the Requirement Review process, delivered in the same data package. All referenced documents are listed at § 2.2.

1.2 SYSTEM OVERVIEW

The ST is a complex system aimed to provide suitable support during countermeasure campaigns testing activities.

It is composed by:

- a Control Unit, allowing to process data and control all device equipment.
- the Turret, hosting the seekers and cameras
- the cover, aiming to protect the device during transport and storage.

The Control Unit

The Control unit will perform following main activities:

- The CU has to receive from the Coral Mainframe the commands/messages needed for
  - FoVs Steering, LoSs Alignment and Lead Angles Application;
  - Cameras control;
  - Turret control;
- The CU has to send to the Coral Mainframe the messages needed for the monitoring of:
  - Steering angles, alignment angles and Lead angles;
  - Turret and Aiming Camera;
- The CU has to integrate a Control Panel in order to guarantee basic manual commands and provide information to the ST Operator;
- The CU has to host and to integrate the Tracking Board “CHARM-100NX” guaranteeing that it is able
  - To receive in input the video frames coming from the Aiming Camera;
  - To make available the tracking results to the CSCI ST Manager;
- The CU has to host and to integrate a board with adequate computing and storage resources for running the ST Manager CSCI.

The Turret

The Turret is designed to deploy eight seekers (SUTs) and two Cameras and to steer their FoVs towards a reference direction aiming flying platform targets. The Turret supports the following SUTs (simultaneously present) configurations:

- 6 SUT “heavy and compact”/”heavy and voluminous”+ 2 SUT “light and compact”;
- 8 SUT “light and compact”;
- 4 SUT “heavy and compact”/”heavy and voluminous” + 4 SUT “light and compact”.

The use of the ST involves:

- Seekers installation;
- Cameras and Seekers alignment;
- Operations to allow seekers calibration with Coral (video marker adjustments);
- Steering of SUTs and Cameras FoVs (Manually or Automatically) and Lead Angles application during target tracking.

The ST can be used independently or integrated with its external manager apparatus called Coral Mainframe.

The main functions of the ST are the following:

- FoV steering, the SUTs and Cameras FoVs are oriented in azimuth and elevation either in manual mode or, on the base of the “Target Tracking” output information, in automatic mode;
- Line of Sight (LoS) alignment, the SUTs and Cameras FoVs are angularly adjusted during the initial phase of their deployment on the ST, at the end of the alignment procedure the LoSs will be parallel each-other;
- Lead Angle Application, The Lead angle is the relative angle between the target direction and the SUT mechanical axis during its operation. Each SUTs has to be oriented in order to impose a Lead Angle;
- ST Control, the ST can be controlled either remotely (coming from either an external user or the Coral Mainframe) or through its Control Panel
- Target Tracking, while tracking the ST controls the azimuth and elevation of SUTs and Cameras FoVs in order to keep the flying target monitored;
- Self-levelling, the ST self-levels within specified range and accuracy on the horizontal direction;
- SUT protection, a suitable cover to protect the SUTs and Cameras from weather and weak intrusions is deployed;
- User Interface, an external user through HMI can visualize the state of the ST and send commands;
- SUTs cables connection, the cables directed to the SUTs are conveniently passed through the ST.

The Housing

The Housing has the function to provide full protection to the Turret from the weather conditions and to allow the safe transport.

1.3 DOCUMENT OVERVIEW

This document is composed by the following sections:

- **Chapter 1** “Identification”, containing the general information on the document and a short description of the system it refers to.
- **Chapter 2** "Reference documents", listing the applicable and referenced documents;
- **Chapter 3** "Design Justification File", describing the main technical solutions and justifications supporting the choices.
- **Chapter 4** "Traceability Matrix", when applicable (not for this issue).
- **Appendices**, containing the appendices (if any).

2 REFERENCE DOCUMENTS

2.1 APPLICABLE

The documents listed in this section are considered to form part of this document. They are only applicable in so far as they are called up by individual sections of this document. For undated references, the latest edition of the document referred to shall be applicable.

Ref.	Document	Issue/rev - Date
[A.01]	Contract n. 4500065039/141	EDOC00288475 rev 1
[A.02]	Statement Of Work for the Steering Turret Coral	EDOC00238664 rev 2
[A.03]	Steering Turret Coral – Control Unit HRS	E0287553 01 rev 3
[A.04]	Steering Turret Coral - Housing HRS	E0287554 01 rev 2
[A.05]	Coral Steering Turret - Turret HRS	E0287561 01 rev 3
[A.06]	Requisiti per I Fornitori ELT	Q0002501 DV rev 12

Table 3 - Applicable documents

2.2 REFERENCE

The documents listed in this section serve as reference material or possible sources of clarification and are for information only. For undated references, the latest edition of the publication referred to shall be consult.

Ref.	Document	Issue/rev - Date
[R.01]	QMS - Manuale di Qualità ALMA	ALMA-QMS-0001 v1.2 09/01/2024
[R.02]	MIL-STD-498	/ /
[R.03]	Steering Turret Coral CAD drawings	ALMA-STC-CAD-011 v1.0 31/10/2024
[R.04]	Interface Control Document	ALMA-STC-ICD-016 v1.0 31/10/2024
[R.05]	Electrical Interconnection Diagram	ALMA-STC-TN-012 v1.0 31/10/2024
[R.06]	CTC – Carrello Elevatore	PL 306-308-310-313-315.-317 3R

Table 4 – Reference documents

3 DESIGN JUSTIFICATION FILE

Following terminology and naming convention will be used in the document:

Turret: with the term Turret, we refer the turret itself, including all motors, mirror supports, nitrogen distribution;

Platform: the term refers to the base of the Turret excluding the turret and the turret motor. The platform includes the physical support, with the self-levelling feet;

Cover: the term refers to the ACU Digital Urban colour cover that must be used to protect the Turret in case of bad weather conditions.

Z-panel: the term refers to the connector panel. The platform will provide a unique panel to connect all wires. The panel will be located on the back side of the platform.

Camera block: the term refers to the head of the Turret, composed by the camera mechanical interface and its motor.

3.1 HOUSING

The housing is designed for easy installation, requiring no special tools. The use of an aluminum frame combined with a polyurethane sandwich for the walls helps to keep the total mass of the cover around 80 kg. The removable walls allow one or two operators to remove the lid from the turret without the need for a crane and without requiring significant height clearance.

On the rear side of the lid, there is an interface for a forklift (see § 3.2). As an option, the lid is equipped with three sets of handles, located on each of the three sides, allowing operators to manually remove the lid when necessary. The lid, excluding the door, weighs approximately 65 kg.

A set of sealing gaskets is mounted at the interfaces between the cover and the baseplate, as well as between the cover and the door. These gaskets ensure that the cover provides effective protection for the turret against weather conditions.

Following Table 5 shows the full size of the housing (**[444580]**).

Height	2352 mm
Width	2403 mm
Depth	2403 mm
Mass	80 kg

Table 5 – Housing size

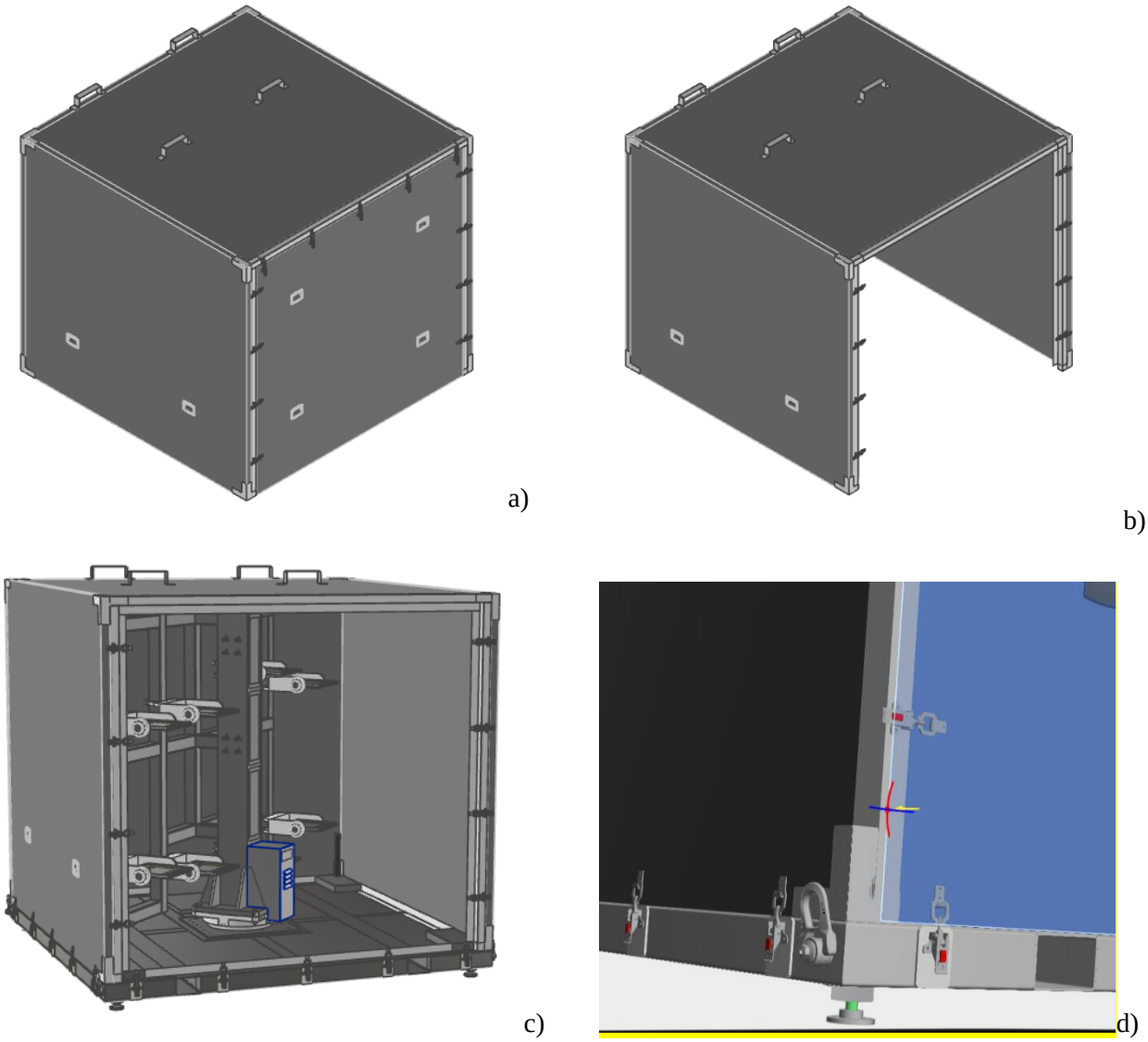


Figure 1 – Housing with fork list on back side (a); the removable panel on front side (b); Open Housing with Turret inside (c); fixing hinges (d)

3.2 MOVING WITH FORKLIFT

CTC[1] forklift catalogue (ref [R.06]) provided by ELT was considered for the following analysis.

The total weight (Turret + Housing) W_{Turret} is estimated around 800 kg.

Considering the residual capacities of the PL306 model (the model with the lowest nominal capacity), it is not possible to guarantee the mobility of the Turret.

We evaluated the capacity needed Q_{min} to move the turret:

- keeping the centre of gravity CG_{PL}= 500 mm for all the PL models as described by the manual
- estimating the weight of the turret W_{Turret} ~ 800 kg
- taking the platform as spacious as possible to keep the balance, putting the CG_{Turret} = 1200 mm

[1] https://ctccarrellielevatori.it/

as $Q_{min}= (W_{Turret} \times CG_{Turret})/CG_{PL}$ = 1920 kg.

This raw estimation excludes the possibility to use the PL models to move the turret. Indeed, the maximum forklift capacity described by the manual is 1200 kg

3.3 CONTROL UNIT

Control unit is assembled in an 12U rack. It is composed of the following units:

The Control Unit is composed by:

- The Control Panel [4U] (see § 3.3.1);
- The CHARM-100NX Frame Grabber card [1U], installed as per requirement.
- Computer [2U], running the ST Manager CSCI (see § 3.3.3). The computer will be equipped with its own graphic board, managing the Control Panel display and a Digital IO board (to manage discrete signals);
- Switch 24 ports, for the internal network and SUTs management;
- Power Supply, to provide different voltage power to CU equipment;
- Power Distribution Unit, to distribute power to CU equipment,

according the disposition shown in the Figure 4.

The main component is the control computer, housed in a rack cabinet. All peripherals are interfaced with the computer, as can be seen in the detailed block diagram. The CHARM 100NX video processing board is responsible for the tracking functions, processing the video data coming from the steering turret aiming camera. The LAN switch/router provide for all the LAN connections required to handle the signals coming from the SUT's and routed to the CORAL mainframe.

The control and power cables coming from the steering turret are attached at the two respective connectors placed in the rear section of the rack. LAN line coming from the Aim camera is routed to the CHARM 100NX video processing board. Aim camera control via the RS-485 bus is provided by the Control Computer. Discrete signals coming from the steering turret (SAFE, STEER, FAIL) are routed to GPIO lines.

Other serial and LAN interfaces are routed to the rear panel, to provide an alternative source for the tracking data (normally provided by the CHARM 100NX) either by serial bus or by LAN interface. This information is generally provided by an external tracking radar.

Control unit acts also as a PDU (Power Distribution Unit), providing the AC 220V to the Steering Turret by means of the main circuit breaker. Two discrete signals enable relays in the steering turret to power on and to arm the turret by enabling the power to the motor actuators.

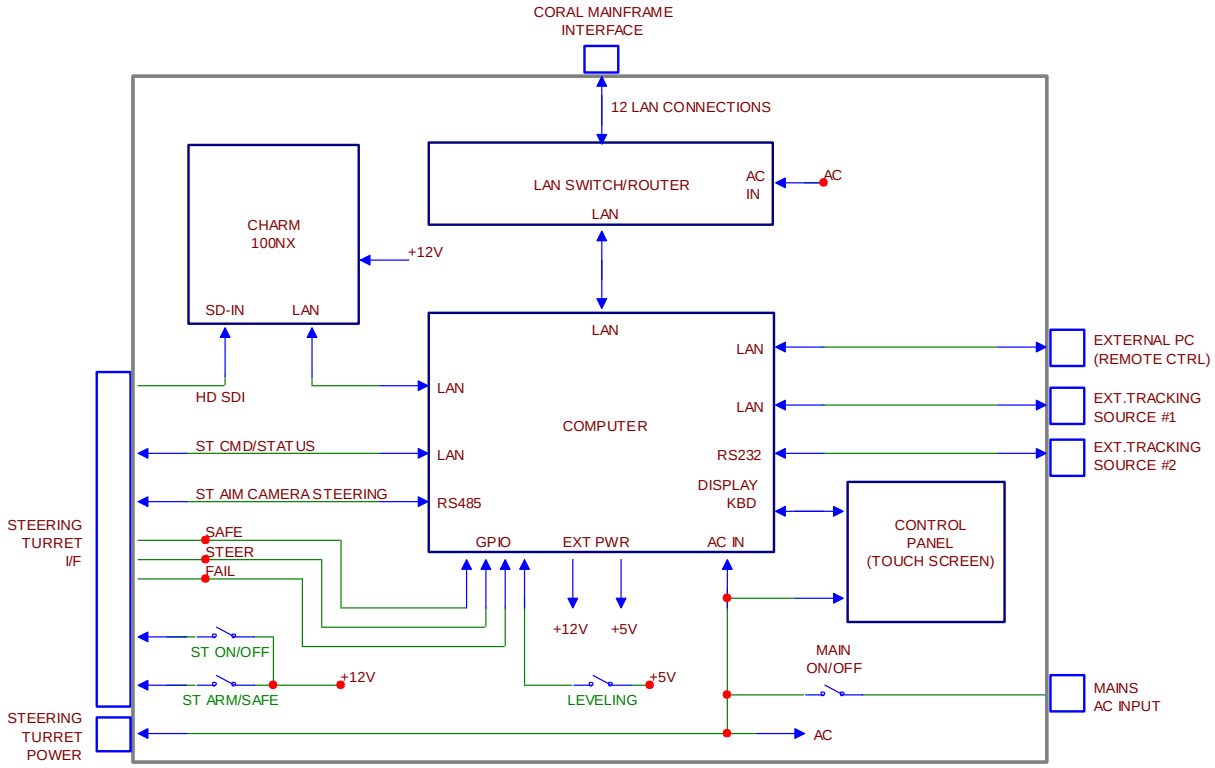


Figure 2 – Control Unit functional block diagram

3.3.1 Control Panel

The Control panel is basically made by a graphic display with touch screen, and by some switches and LED indicators, as follows:

- MAIN ON/OFF – circuit breaker, which powers ON the Control Unit, and delivers AC 220V to the Steering Turret circuits.
- ST ON/OFF – switch, which powers ON the Steering Turret.
- ST ARM/SAFE – switch, which puts in SAFE mode, or ARMs the Steering Turret motor actuators.
- LEVELING – Push button, which starts the levelling function of the Steering Turret.

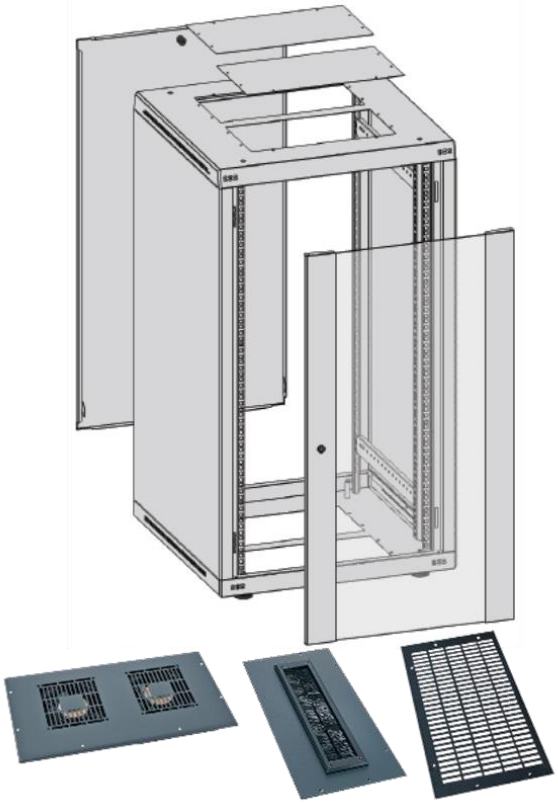
A series of indicators will show the operating status of the Steering Turret, and of the levelling function. Other functions are available by means of the touch screen.

3.3.2 Rack assembly

The control unit will be hosted by a 12U rack, composed as described in the next Figure 3.

The MRAK® 610 is a medium-sized floor standing cabinet, particularly suitable for housing networking equipment and 19” rack equipment in industrial plants. It is designed to fit into limited work surface spaces. IMRAK® 610 features a welded structure that ensures a high degree of robustness and protection with a static load capacity of 250 kg. Reversible front doors are available with a central key lock. The removable rear panel is equipped with a quick lock for easy opening and closing. The side panels are fixed. Cable entry is possible from the base or from the ventilated top cover. A two fans panel will be installed in the roof.

The 19” uprights are adjustable in depth and allow for the housing of active equipment of different sizes or for having a larger front space for the installation of interconnection panels.



Cabinet **IMRACK 610 19'' "PRIMARY"**

Internal height: 12U

External height: 634 mm

Depth: 600 mm

Lenght: 600 mm

Weight: 15 Kg

Load capacity: 250 Kg

Removable back panel

Glass front door

Upper / lower cable entry brush kit (partial)

Ventilation group: 2 low noise fans (partial)

Ventilated cover kit

Kit guides 470 mm

Kit top 370 mm (cabinet depth 600)

Earthing kit for mobile cabinet parts

Multisocket rack 19″, 7 sockets UNEL BIPASSO, 16A, magneto termico 2P+N, cable 2 mt, plug Shuko

1U 19″ copper ground bar kit with insulators

Figure 3 – CU rack layout

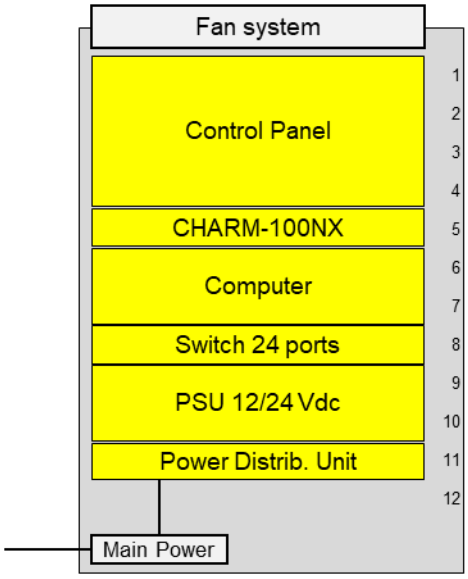


Figure 4 – CU equipment disposition

3.3.3 Software (ST Manager CSCI)

3.3.3.1 High-Level Functionality

The CU is designed to fulfill a range of functionalities, ensuring flexibility in its operation and control:

- **Communication Management**: The CU facilitates communication between the Coral Mainframe, External Trackers, and the turret, employing specific protocols for sending and receiving commands, telemetry data,

and status updates. This involves interpreting commands related to Field of View (FoV) steering, Line of Sight (LoS) alignment, turret control, and lead angle applications.

- **Command Processing and Execution**: The CU receives commands from the Coral Mainframe and other input sources to manage turret operations, Field of View (FoV) steering, Line of Sight (LoS) alignment, and lead angle applications. These commands are executed in real-time, ensuring the system responds accurately to operator inputs.
- **Real-Time Data Monitoring**: The CU sends monitoring data back to the Coral Mainframe, which includes telemetry related to turret position, alignment angles, and overall system health.
- **Operational Mode Handling**: It supports multiple operational modes, including Calibration, Manual, and Automatic. Each mode caters to different aspects of the system's operation:
  - **Calibration Mode**: This mode focuses on configuring and storing angular alignments and lead angles for both the turret and cameras.
  - **Manual Mode**: Allows the operator to manually control the turret's movement and aiming using commands received from the Coral Mainframe.
  - **Automatic Mode**: Supports autonomous control based on input from the Aiming Camera or external tracking sources, allowing for precision steering and tracking of targets.
- **Built-in Test (BIT) Execution**: The CU is equipped with BIT functionalities (PBIT, CBIT, and IBIT) that allow self-diagnosis to verify equipment health during various states, such as startup, periodic checks, and on-demand operations.
- **Safety and State Management**: The CU manages a comprehensive set of operational states—INIT, FAIL, MAINTENANCE, OPERATIONAL, and SAFE—ensuring the system responds appropriately to various conditions, commands, and failure scenarios
- **Control Panel Integration**: The CU is equipped with a Control Panel that provides the operator with a hands-on interface to start/stop turret activities, set safety statuses, control the Self-Levelling procedure and monitor system conditions such as turret movement and levelling progress. The Control Panel communicates with the CU to ensure the operator has real-time control over critical functions.

3.3.3.2 Interaction with External Components

The CU communicates with various external systems and devices through multiple interfaces is depicted in the image below (Figure 5), highlighting the CU's central role in controlling the turret, receiving commands, and managing communication with external systems. The image outlines the flow of control signals and video data, as well as the different interfaces (Ethernet, RS-485, and HD-SDI) used to connect the CU with the Coral Mainframe, tracking sources, Aiming Cameras and the turret itself.

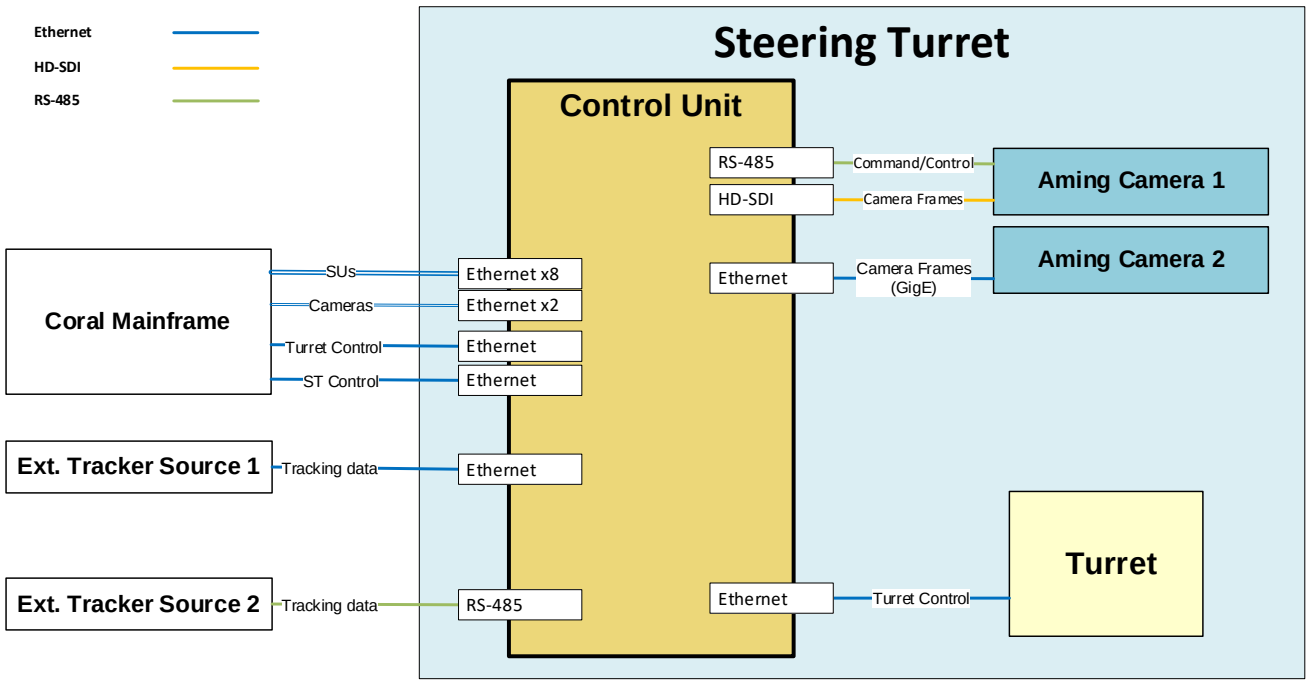


Figure 5 - Control Unit External Connections

Each external component serves a specific role in the overall system architecture:

- **Coral Mainframe**: The CU communicates with the Coral Mainframe over Ethernet connections to exchange control commands and system telemetry. Commands from the Coral Mainframe include steering and alignment control for the turret and Aiming Cameras. The CU will adopt the communication protocol already defined.
- **Aiming Cameras**: Aiming Cameras are connected to the CU through RS-485 and HD-SDI line or through ethernet for GigE cameras. The RS-485 interface handles camera control commands, while the HD-SDI lines transmit camera frames for automatic tracking and target identification. For GigE cameras the control communication and grabbed frames will use the ethernet connection. CU implements control and monitoring protocols provided by the manufacturer for the specific camera.
- **External Tracking Sources**: The CU supports two external tracking sources. These sources provide tracking data, which can be integrated with the CU's control systems to enhance automatic turret tracking capabilities. Ethernet and RS-485 interfaces are used for communication. Communication protocols to be defined.
- **Turret**: The turret is directly controlled by the CU, with Ethernet used for sending movement commands and receiving monitoring data. The CU ensures that all turret movements are within safe angular limits, and it can saturate commands if necessary to avoid exceeding these limits. Communication protocol to be defined.

3.3.3.3 Interaction with Internal Components

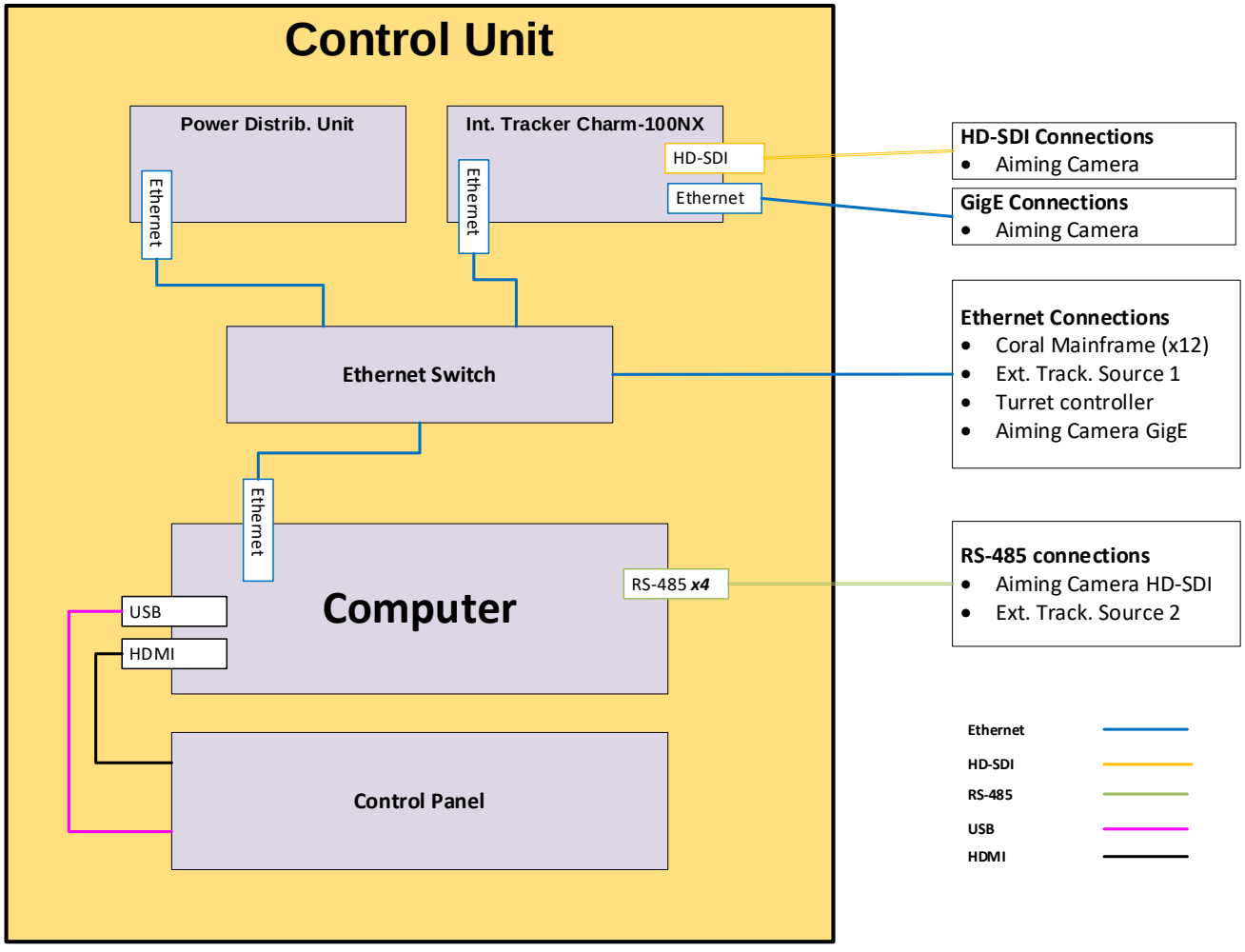


Figure 6 - Control Unit External Connections

The internal components of the CU include the computing unit, the Control Panel, the CHARM-100NX tracking board, and the Power Distribution Unit (PDU). These components interact closely to ensure seamless system operation.

- **Computing Unit**: The embedded computing unit runs the CU software and processes real-time data for controlling Field of View (FoV) steering, Line of Sight (LoS) alignment, Aiming Camera control, and turret operations. It is responsible for coordinating all internal data flow and executing commands received from external sources.
- **Control Panel**: The Control Panel allows the operator to interact directly with the system, providing manual control, system checks, and state transitions. The Control Panel is installed in the same rack as the rest of the CU instrumentation. It contains commands for Power ON and ARM/SAFE of the Turret, as well as LEDs to indicate the status of these commands. The Control Panel is also equipped with a touchscreen display, connected via HDMI/USB, which allows the operator to initiate the Self-Levelling procedure and view the system status in real-time. Additionally, the Control Panel integrates a USB Digital I/O interface for monitoring the SAFE/ARM status and other digital signals. This integration provides a user-friendly interface for both monitoring and controlling key aspects of the system, ensuring smooth operation and safety.
- **CHARM-100NX Tracking Board**: The CHARM-100NX is integrated for advanced tracking capabilities. It processes input from the Aiming Cameras and works in tandem with the computing unit to ensure precise turret positioning and automatic target tracking. Depending from the type of camera the grabbed frames will be transferred to this unit through the HD-SDI or GigE interface.
- **Power Distribution Unit (PDU)**: The PDU manages power distribution across the CU, ensuring stable and reliable power supply to all internal components. It interacts with the computing unit to monitor power status and report any anomalies.

These components are interconnected through an Ethernet connection that facilitates the exchange of telemetry, operational commands, and sensor data. Externally, the CU interfaces with the Coral Mainframe and Aiming Cameras to receive commands and send system status updates, ensuring coordinated operation of the entire turret system.

3.3.3.4 State Machine Overview

The Control Unit (CU) manages the turret's operation using a state machine with five primary states: **OFF**, **INIT**, **SAFE**, **OPERATIONAL**, **MAINTENANCE**, and **FAIL**. The transitions between these states are governed by inputs from the Control Panel and Coral Mainframe commands, internal system diagnostics, and automatic processes. The state machine ensures that the turret operates safely, executes tasks effectively, and responds appropriately to system failures.

3.3.3.4.1 State Machine Diagram

The following diagram illustrates the state machine, showing transitions driven by operator commands trough the Control Panel inputs or from commands received from Coral Mainframe and automatic transaction system due to check and operation results (PBIT, CBIT, IBIT).

- **Orange Arrows**: Indicate transitions triggered by the Control Panel (operator commands).
- **Blu Arrows**: Indicate transitions triggered by commands received from Coral Mainframe (operator commands).
- **Green Arrows**: Indicate automatic transitions based on internal system diagnostics (PBIT, CBIT, IBIT).

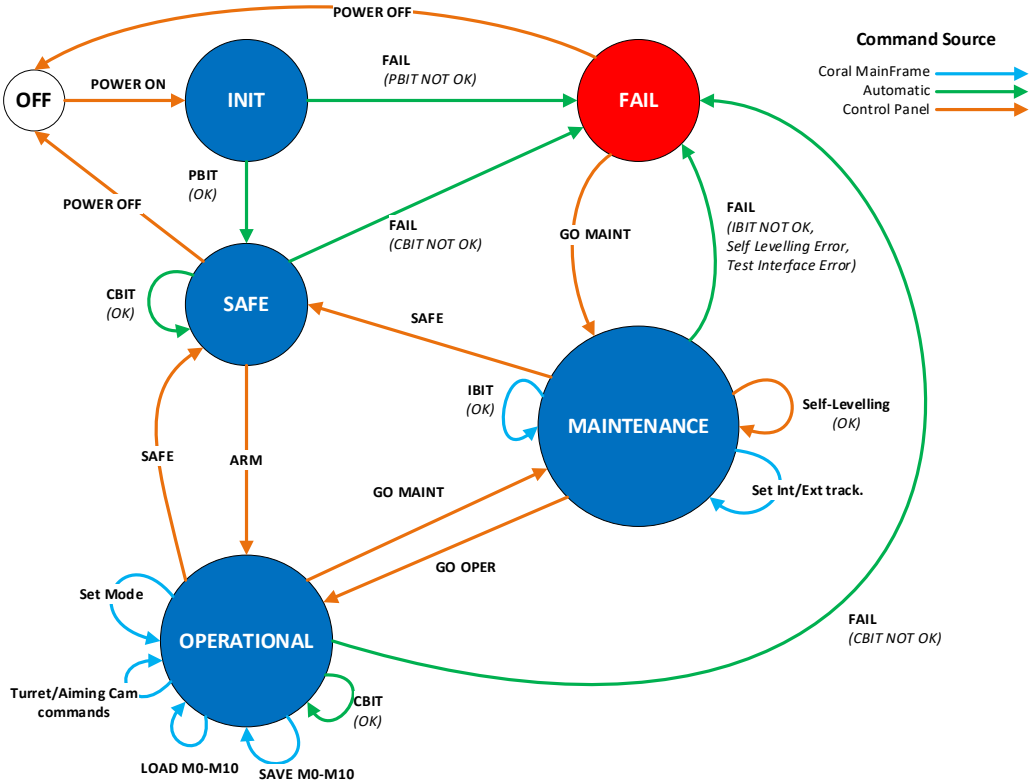


Figure 7 - Control Unit State Machine and Commands

3.3.3.4.2 State Descriptions

- **OFF State**: the system begins in the OFF state when it is not powered, and no operations occur. The CU enters this state when the system is turned off via the Control Panel. No actions are performed in this state, and no components of the system are active.
- **INIT State**: upon powering on the system, the CU enters the **INIT** state. In this state, the system performs a Power-On Built-In Test (PBIT) to check the health of critical components, see Table 20 for the detail. The system remains in the **INIT** state until the PBIT confirms that all systems are functioning correctly. If the PBIT passes, the system transitions to the **SAFE** state. If it fails, the system enters the **FAIL** state.
- **SAFE State**: the **SAFE** state ensures that the turret is locked, with no movement allowed. Transitions to and from **SAFE** can only be controlled via commands issued from the Control Panel. Upon successfully completing the PBIT, the system transitions to **SAFE**, awaiting further commands. From the **SAFE** state, the only permitted transition is to the **OPERATIONAL** state, based on a Control Panel command *ARM*. Similarly, the system can return to **SAFE** from the **OPERATIONAL** state when the operator issues a *SAFE ON* command from the control panel.
- **OPERATIONAL State**: the **OPERATIONAL** state is where the turret performs its intended functions. In this state, the CU manages the following modes:
  - *Calibration*, in which CU allows to define angular alignments of the SUTs and Cameras and the lead angles to the SUTs defined by the user;
  - *Manual*, in which the CU allows the user to manually control turret azimuth;
  - *Automatic*, in which the ST steering the FoV automatically on the base of the Aiming Camera output or the output of one of the two possible external tracking sources.

  The transition to **OPERATIONAL** occurs when the Control Panel issues a *ARM* command from the SAFE state or the remote command *GO OPER* from Coral Mainframe. The system remains in the OPERATIONAL state until it either receives a *SAFE ON* command from the Control Panel, receives a *GO MAINT* command which brings the system in **MAINTENANCE** state, or a failure is detected, which forces

the system to transition to **FAIL**. Every time the CU enters the **OPERATIONAL** state it automatically set Mode to Manual.

- **MAINTENANCE State**: the **MAINTENANCE** state is used for diagnostic, external tracking configuration and self-levelling activities. The system enters this state when from OPERATIONAL it receives the remote command *GO MAINT*. In this state, the system supports activities such as turret self-levelling, IBIT checks, automatic tracking configuration settings and camera configuration settings. After completing maintenance, the system can transition back to **OPERATIONAL** or **SAFE**, depending on the operator's commands. If errors occur during maintenance (e.g., IBIT failure, self-levelling issues), the system transitions to the **FAIL** state.
  If the system is in the FAIL state, it can transition back to the **MAINTENANCE** state upon receiving a GO MAINT command. This transition allows for analysis of the failure and the execution of verification tests to diagnose and resolve the issues.
- **FAIL State**: the **FAIL** state is triggered when critical errors occur that prevent the turret from operating safely. These errors can be detected during PBIT, CBIT (Continuous Built-In Test), IBIT (Initiated Built-In Test), or during operational or maintenance activities (e.g., self-levelling failure). In the **FAIL** state, the system halts all operations and the operator can switch off the system or, trough the Control Panel, set the system to **MAINTENANCE** state for verifications and troubleshooting.

The Table 6 outlines the key commands used to interact with and control the CU system during its different states. These commands can be sourced from the CU itself, Control Panel, Coral Mainframe, and are processed by the CU for actions such as turret movement, mode selection, or system diagnostics.

The command set supports full control of the CU's capabilities, enabling safe operation, real-time diagnostics, and flexible configuration based on mission requirements.

Commands	Description	Source
Power ON	Power On the Control Unit	Control Panel
Power OFF	Power Off the Control Unit	Control Panel
PBIT	Power on Built-In test	Automatic
CBIT	Continuous Built-In test	Automatic
ARM	Arms the motor	Control Panel
SAFE	Power off the motors	Control Panel
Set Mode	Set the Operational Mode (Calibration, Manual, Automatic)	Coral Mainframe
Turret/Aiming Cam commands	Commands to configure Turret and Aiming camera	Coral Mainframe
SAVE M0-M10	Save the current config to one of the available memory	Coral Mainframe
LOAD M0-M10	Load config from one of the available memory	Coral Mainframe
GO MAINT	Put the system in MAINTENANCE state	Control Panel
GO OPER	Put the system in OPERATIONAL state	Control Panel
IBIT	Run the specified Initiated Built-In Test	Coral Mainframe
Set Int/Ext track	Set the source of the automatic tracking	Coral Mainframe
Self-Leveling	Execute the Self-Leveling operations	Control Panel
FAIL	Puts the system in FAIL state on any critical error	Automatic

Table 6 - Commands

3.3.3.5 Built-In Tests

The Control Unit (CU) integrates three types of Built-In Tests (BIT) that ensure the system's readiness and reliability across different operational phases:

1. **Power-On Built-In Test (PBIT)**: Conducted at system startup while the CU is in the INIT state, the PBIT checks the health of essential system components such as the Aiming Cameras, the CHARM-100NX tracking board, the Power Distribution Unit (PDU), and the Turret. PBIT is critical to verify that all primary components are functional before proceeding to the SAFE or OPERATIONAL states. Results are consolidated and reported to the Coral Mainframe for confirmation.

   The PBIT must complete within a predefined timeout period; if the timeout is exceeded, the system transitions to the FAIL state.

2. **Continuous Built-In Test (CBIT)**: CBIT operates periodically during the SAFE and OPERATIONAL states. The CBIT monitors key system parameters, such as health status of Aiming Cameras, the Turret Controller program, motor health, and the interface between the CU and the Turret. This test ensures any emerging failures are quickly identified and mitigated to minimize potential downtime and safety risks.

   CBITs are executed in one or more separate threads, each with its own frequency depending on the criticality of the parameter being monitored. For example, motor health checks might be performed every 100 milliseconds to ensure rapid detection of issues, while system temperature checks might be executed every 5 seconds. If the CBIT does not complete within its timeout period, the system will transition to the FAIL state.

3. **Initiated Built-In Test (IBIT)**: IBIT is performed during MAINTENANCE operations, either automatically or upon request from the operator through the Control Panel. The IBIT ensures that all calibration and diagnostic activities are performed accurately. The IBIT covers detailed checks such as angular limit tests and turret configuration assessments. The results are used to validate the system's operability before resuming standard operations.

   The IBIT must also complete within a defined timeout period; failure to do so will result in the system entering the FAIL state.

These BITs collectively offer a thorough mechanism for verifying system health during different stages of operation, minimizing risks associated with potential faults.

The BIT table in Table 20 shows tests performed on different system components. The columns 'CU' and 'Turret' indicate which parts are covered by each BIT. The CU column covers tests on the Control Unit, including software and interfaces. The Turret column covers tests on mechanical and control elements, such as motor health and angular limits.

3.3.3.6 High-Level Software Architecture

The Control Unit (CU) software is based on a Linux OS (long-term support version) for reliable real-time operations. C++ is used for its efficiency and hardware control, crucial for turret control and communication

The architecture is modular, with separate software components responsible for different functionalities, such as communication with external devices, system diagnostics, control of the turret, and the user interface on the Control Panel. Each module interacts with others via well-defined interfaces, ensuring scalability and maintainability. Real-time processing is crucial for handling commands and telemetry data, ensuring that the turret responds promptly to operator inputs and external conditions. The system also includes a monitoring module that continuously checks the health of both hardware and software components, providing feedback for operational decisions. Additionally, the software includes a detailed event log that records all significant actions and state changes, supporting diagnostics and traceability.

Software Modules

The CU software is divided into several key modules:

1. **Communication Modules**: Multiple communication modules manage communication between the CU and external systems:
   - **Coral Mainframe Communication Module**: Handles data exchange between the CU and the Coral Mainframe.

- **External Tracker Source 1 Communication Module**: Interfaces with the first external tracking source.
- **External Tracker Source 2 Communication Module**: Interfaces with the second external tracking source.
- **Turret Controller Communication Module**: Manages communication with the turret controller, ensuring that commands for movement and feedback are reliably transmitted.
- **Aiming Cameras Communication Module**: interface the Aiming cameras for monitoring and configuration.

These modules handle the transmission and reception of commands and telemetry over Ethernet and serial connections, ensuring reliable data exchange with other components.

2. **Control Module**: Responsible for executing commands to control the turret, including Field of View (FoV) steering, Line of Sight (LoS) alignment, and lead angle adjustments. This module interacts with the turret controller, providing precise control based on operator or automatic commands. It also manages the state machine, ensuring that the CU transitions between states (INIT, SAFE, OPERATIONAL, etc.) based on system conditions and commands received.
3. **Diagnostic Module**: Implements the Built-In Test (BIT) functionalities, including PBIT, CBIT, and IBIT. It continuously monitors the health of the system, performs diagnostic tests, and reports any faults to the operator, ensuring system reliability.
4. **User Interface Module**: Manages the touchscreen Control Panel, providing the operator with real-time system status information and control options, such as Self-Leveling commands. This module also monitors input from the GPIO board to track the SAFE/ARM status of the turret.
5. **Monitoring Module**: Continuously monitors system parameters, such as power distribution, temperature, and component health, providing feedback to the control and diagnostic modules. This module is critical for maintaining operational safety and stability.
6. **Logging Module**: Maintains a detailed event log of all system actions, commands, and state changes. This module ensures traceability and aids in diagnostics by providing a historical record of system activities. Log files will be kept according to dedicated space. Old files will be automatically deleted if needed.
7. **Tracking Management Module**: Manages the CHARM-100NX tracking board, which processes input from the Aiming Cameras to ensure precise turret positioning and automatic target tracking. It works closely with the Control Module to provide accurate data for turret control during automatic tracking operations.
8. **Control Panel Management Module**: Manages the Control Panel, including the touchscreen interface connected via HDMI/USB. This module is responsible for displaying system status and allowing the operator to execute commands, such as Self-Leveling. It also interfaces with the GPIO board to monitor digital signals, like the SAFE/ARM status of the turret, and ensures that operator inputs are processed effectively.
9. **Configuration Management Module**: Manages settings like IP addresses, ports, data storage paths, and logging options. This module maintains configurations and provides an interface for updates, ensuring consistency and smooth operation across the system.

3.3.3.7 Operational Modes

3.3.3.7.1 Calibration

The Calibration Mode allows the Control Unit (CU) to receive Line of Sight (LoS) alignment values for each Sensor Under Test (SUT) from the Coral Mainframe and save them in memory location M0. Additionally, the CU receives Lead Angles for each SUT, which are then stored in memory locations M1 through M10.

3.3.3.7.2 Manual

The Manual Mode allows the operator to take direct control of the turret and aiming cameras. Commands are received from the Coral Mainframe, or with a joystick for easier manual operation.

In Manual Mode, the CU supports Field of View (FoV) steering, which is manually controlled by the Coral Mainframe, as well as turret and aiming camera control. The default configuration for Line of Sight (LoS)

alignment angles (azimuth and elevation) is loaded from memory M0 for each SUT and Turret. The CU can also load and apply angles from memory locations M1-M10, which contain specific lead angle configurations for each SUT.

This is the default Mode when CU enters Operational state.

3.3.3.7.3 Automatic Tracking

The Automatic Mode allows the CU to autonomously control the turret and SUTs using either internal or external tracking sources. In this mode, the CU continuously adjusts the turret's orientation and the SUT angles to keep the target centred. Upon entering this mode, the values saved in memory M0 are automatically loaded. The CU uses tracking data from either an internal tracking board or an external source.

3.3.3.7.3.1 Internal tracking source

Based on the requirements and recent discussions, we can outline the following usage scenario for the system when using the internal tracking source:

The turret is positioned so that the target's trajectory falls within its rotation range of -140° to 140°. Before the test begins, the turret is rotated to ensure the aiming camera is pointing at the target's starting point. Once the test starts, the turret rotates to keep the target centred horizontally in the camera's frame. During tracking, in addition to turret rotation, the SUT control motors are also adjusted to ensure they are all pointing at the target.

In Figure 8 shows the camera frames at two different moments during the tracking of a target. The figure illustrates the target's trajectory (blue dashed line), the path taken (red line), the target (red cross), the camera frame (green rectangle), and the horizontal midpoint of the camera frame (yellow line).

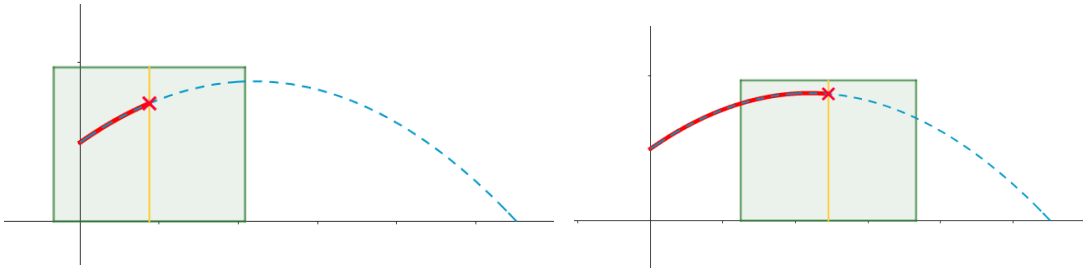


Figure 8 – Camera Tracking

The CHARM-100NX tracking board must be configured with the aiming camera's Field of View (FoV) to provide accurate angles relative to camera boresight.

The CU uses data from the tracking board (Tracked Target to Boresight Error in Milliradians) to rotate the turret and keep the target centred. It also calculates the azimuth and elevation angles for each SUT to keep them aimed at the target (SUT azimuth remains constant for fixed target distance).

Azimuth and elevation calculations must consider parallax error due to the mirrors distance from the camera's line of sight. The Parallax error depends on the distance between the mirror centre and the line of sight of the aiming camera and the distance from the turret to the target. It can be calculated using the following formula:

$$parallax\ error = \arctan\left(\frac{D}{d}\right)$$

Where **D** is the distance from the mirror centre to the aiming camera's Line of Sight, and **d** is the distance from the turret to the target.

The Lead Angles represent the corrections to be applied to the azimuth and elevation of each SUT to compensate for the parallax error

Upon entering this mode, the CU configures the tracking board using settings received from the Coral Mainframe. The tracking management cycle is then started as follows:

- The tracking board uses the connected camera to detect and track the target based on the given configuration.
- The CU receives tracking information from the tracking board to maintain the target lock.
- The CU sends the turret controller new values for turret azimuth and SUT elevation, adjusted according to the Lead Angles from the selected memory.

3.3.3.7.3.2 External tracking source

When using an external tracking source, the CU uses data received from the external source, which must include information about the source's position and orientation relative to a common reference. The CU recalculates the turret azimuth and SUT elevation angles adjusted according to the Lead Angles from the selected memory.

3.3.3.8 Camera Analysis

Some proposed cameras were analysed to verify their compatibility with system requirements. The following table lists the analysed cameras and their characteristics:

Marca	Nome	Focal Lenght mm	res X	res Y	pitch µm	sensor size X mm	sensor size Y mm	H FOV °	V FOV °
FLIR	SC7600	200	640	512	15	9.6	4.92	2.75	1.41
FLIR	SC7600	200	640	512	15	9.6	4.92	2.75	1.41
FLIR	x6580sc	100	640	512	15	9.6	4.92	5.50	2.82
InfraTec	9400	200	1280	1024	10	12.8	13.11	3.67	3.75
InfraTec	8800	200	640	512	15	9.6	4.92	2.75	1.41

Table 7 – Selected Cameras

The compatibility analysis focused on verifying whether the camera/optics combination provides adequate resolution and frame rate for target tracking at specific distances. The following tables show the frame characteristics at two distances: 500 and 5000 meters.

The parameters analysed include:

pixel resolution @distance (mm)	Resolution of single pixel at selected distance
Target size in pixels	Size of the target in pixels at the selected distance
Frame resolution X @distance (mm)	Horizontal dimensions of the framed image in mm
Frame resolution Y @distance (mm)	Vertical dimensions of the framed image in mm
min speed (m/s) @100Hz frame rate	Minimum speed (m/s) the target should have to exit the frame (starting from the centre of the frame) before the next frame is acquired at a frame rate of 100Hz
min speed (km/h) @100Hz frame rate	same as above but in km/h

Table 8 – Camera parameters analysed

Target distance	500 m
Target size	16 m

Marca	Nome	pixel resolution @distance (mm)	Target size in pixels	Frame resolution X @distance (m)	Frame resolution Y @distance (m)	min speed (m/s) @100Hz frame rate	min speed (km/h) @100Hz frame rate
FLIR	SC7600	37.5	427	24	19.2	1200	4320

FLIR	SC7600	37.5	427	24	19.2	1200	4320
FLIR	x6580sc	75	213	48	38.4	2400	8640
InfraTec	9400	25	640	32	25.6	1600	5760
InfraTec	8800	37.5	427	24	19.2	1200	4320

Table 9 – Selected Cameras @ 500 m

Target distance	5000 m
Target size	16 m

Marca	Nome	pixel resolution @distance (mm)	Target size in pixels	Frame resolution X @distance (m)	Frame resolution Y @distance (m)	min speed (m/s) @100Hz frame rate	min speed (km/h) @100Hz frame rate
FLIR	SC7600	375	43	240	192	12000	43200
FLIR	SC7600	375	43	240	192	12000	43200
FLIR	x6580sc	750	21	480	384	24000	86400
InfraTec	9400	250	64	320	256	16000	57600
InfraTec	8800	375	43	240	192	12000	43200

Table 10 – Selected Cameras @ 5000 m

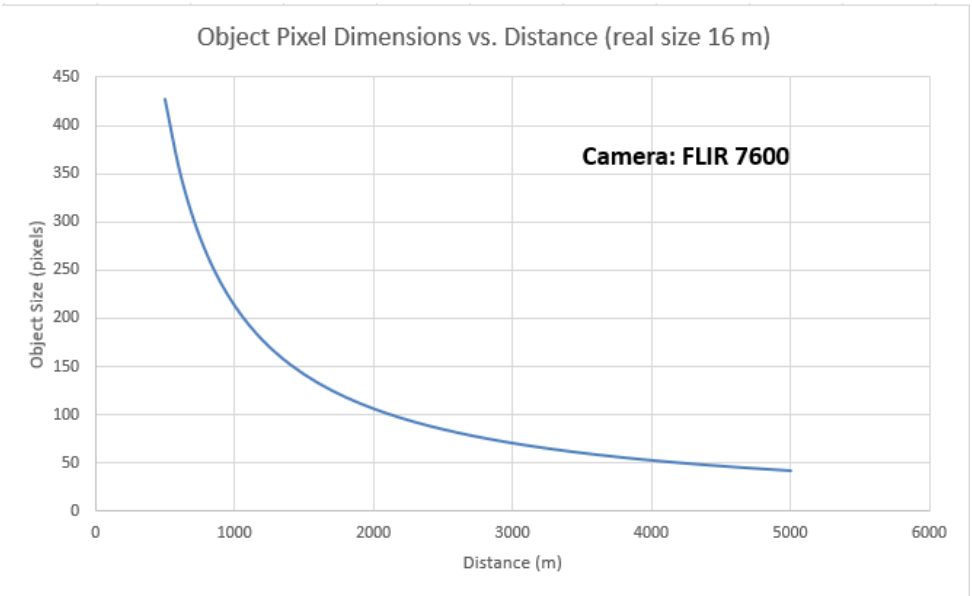


Figure 9 - Object Size On sensor (pixels) at Varying Distances (Camera: FLIR 7600, Object Size 16m)

The graph in Figure 9 shows the pixel size of a 16-meter object captured by a FLIR 7600 camera at distances ranging from 500 to 5000 meters.

In both cases, the resolution and framerate are adequate for tracking, both in terms of pixels allocated to the target and speed requirements. Min speeds must be compared to maximum rotation speed of the turret. The maximum rotation speed of the turret is 15°/s which allows it to follow a target moving at speeds of up to 96 km/h (at 100 m) and 289 km/h (at 300 m).

3.4 TURRET

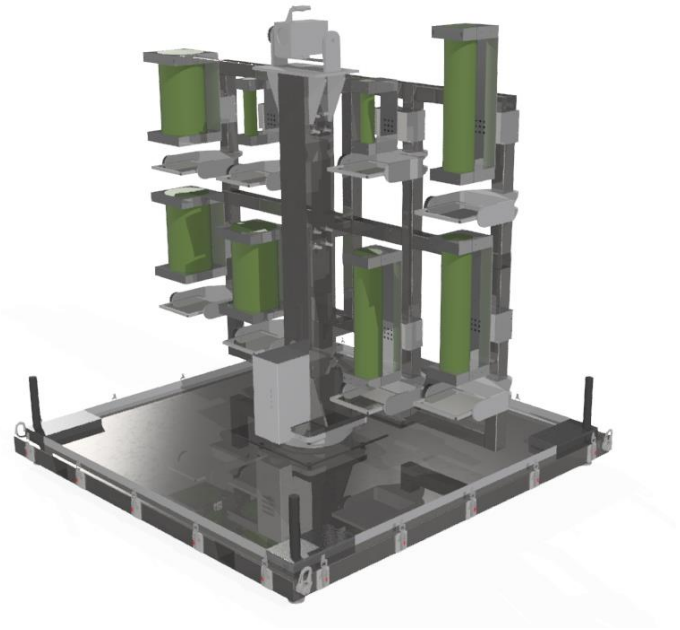
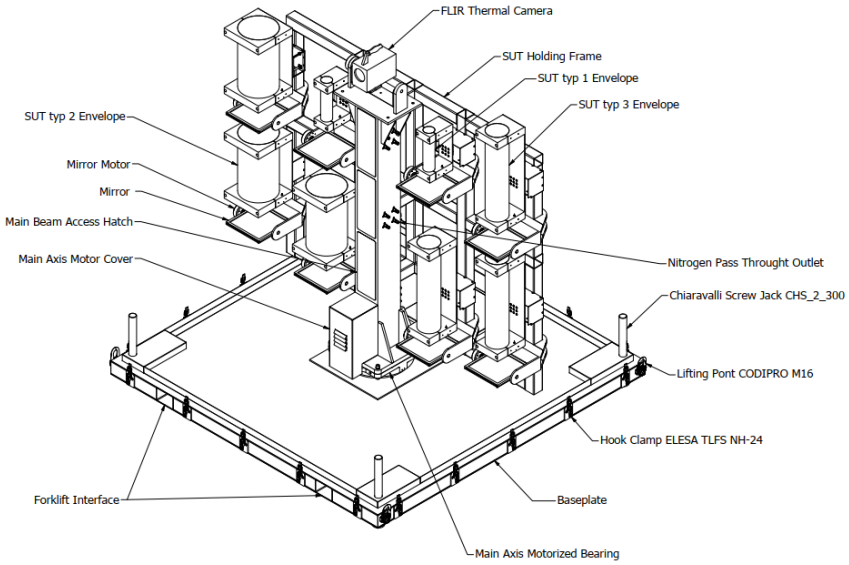


Figure 10 – Steering Turret Overall

The steering turret primarily consists of a square baseplate and a rotatable central beam. Attached to the central beam is a rectangular frame constructed from aluminum profiles, which supports all the SUTs. The baseplate measures 2350 x 2350 mm and is fabricated from welded stainless steel pipes, ensuring strength and durability.

The key components of the turret are detailed in the diagram below:



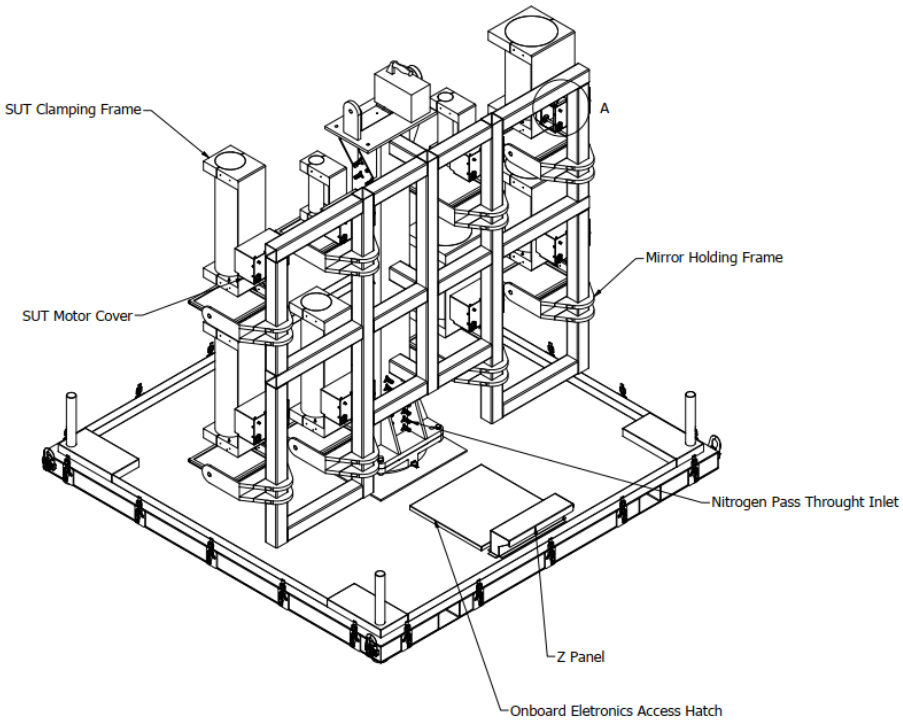


Figure 11 – Steering Turret Components

3.4.1 SUT Adjustment System

Each SUT support has a mechanical four degree of freedom (4 DOF) adjustment system to allow the pupil fine alignment during the calibration:

- Vertical adjustment with micrometric screw:

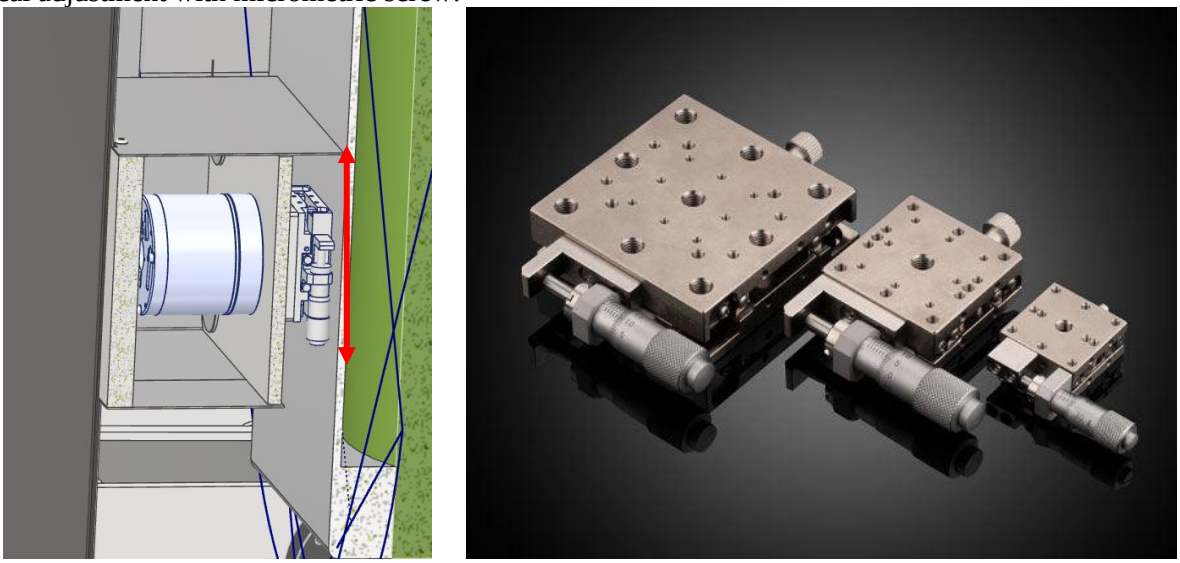


Figure 12 – Micrometric screw

The whole SUT can be moved up and down of 13mm with a micrometric screw with an accuracy of 1µm thanks to a stainless steel extended contact ball bearing linear translation stages.

- 3 DOF screw adjustment system for motor support. This allows two rotation (yaw and pitch) and one linear adjustment:

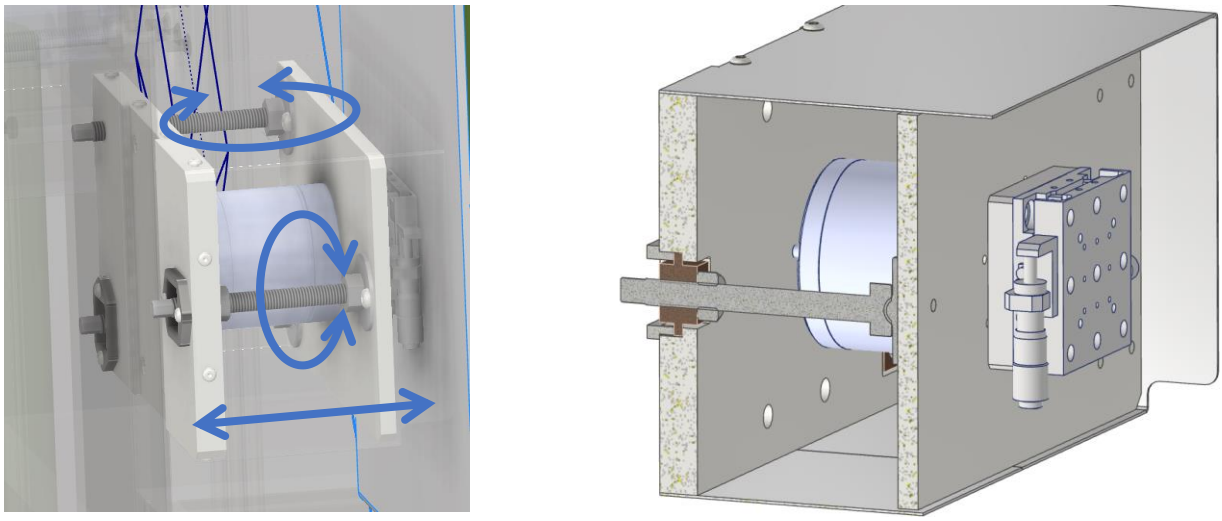


Figure 13 – 3 DOF adjustment

This is possible due to 3 screws connected with three spherical joints on the motor flange. Each screw is mounted in a different way, a fixed one, a yaw orientable nut and a pitch orientable nut; so, each screw will sort a single effect on the motor orientation and moving all three will translate the motor flange.

The whole 4 DOF adjustment system is built in anodized aluminium or aged stainless steel in order to avoid any issue with the oxidation, moreover the system is covered with a metal sheet to avoid direct exposure to atmospheric agents.

3.4.2 SUT Holding Frame

The SUT holding frame is constructed using 90x90L Bosch Rexroth aluminum profiles, chosen for their excellent combination of mechanical performance, dimensional tolerance, and versatility. These profiles not only ensure a robust and reliable structure but also provide compatibility with a wide range of readily available accessories, making it easier to customize and adapt the frame to specific needs. Aluminum material offers strong resistance to oxidation, enhancing the durability of the frame in various environments, while its lightweight nature helps keep the overall mass of the structure relatively low, at approximately 40 kg.

An additional advantage of the 90x90L profile is its ability to accommodate internal cable routing, reducing external clutter and improving system organization. This feature is particularly beneficial for managing the cabling required for the SUT motors, mirror motors, and laser systems, all of which are planned to be integrated within the holding frame. This design not only enhances the aesthetic and functional appeal of the frame but also ensures improved protection for the cables during operation.

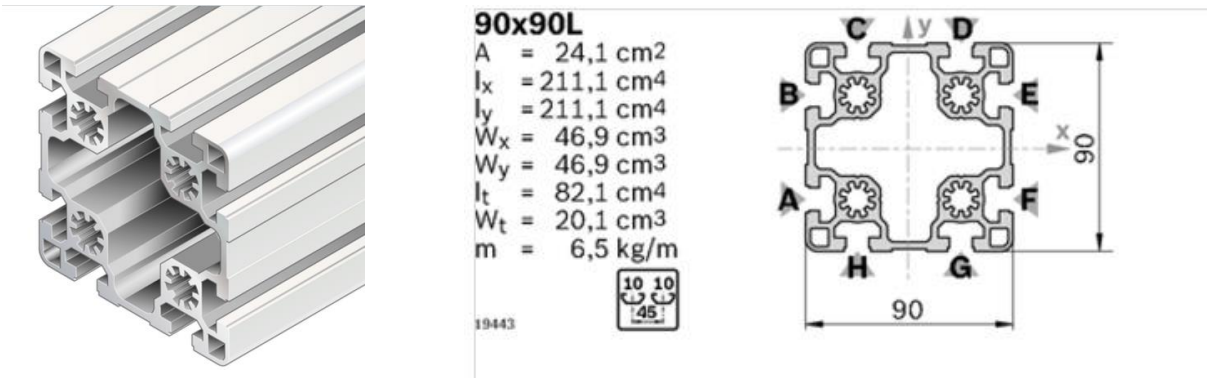


Figure 14 – Bosch Rexroth 90x90L profile

3.4.3 Main Beam

The main beam is constructed from a 200x200x4 mm extruded aluminum pipe (dimensions to be confirmed), with a height of 1900 mm. To facilitate the installation of nitrogen pipes and electrical cables, the beam includes three bolted hatches. These hatches not only simplify the initial setup but also provide quick and convenient access for maintenance and service activities.

A welded flange at the base of the beam serves as the interface for the motorized bearing, ensuring a secure and stable connection. At the front of the beam, a metal sheet cover houses the motor for the bearing, offering protection and easy accessibility. At the rear, a dedicated inlet accommodates the nitrogen feedthrough, while the side of the beam features outlets for the nitrogen feedthrough and the SUTs' electrical connections (see § 3.4.4 and § 3.4.7).

This design combines structural integrity with practical accessibility, enabling efficient operation and straightforward servicing.

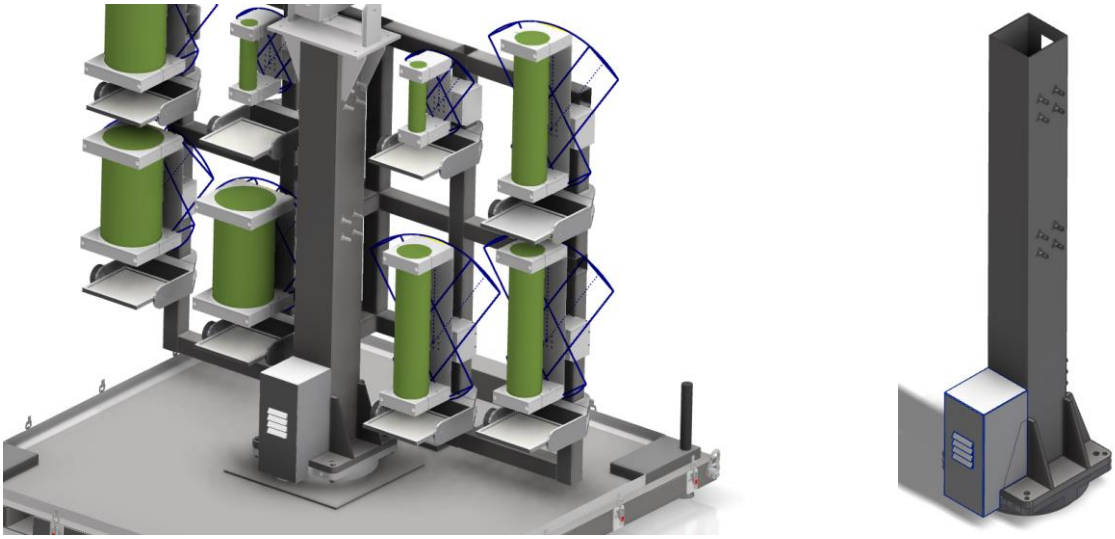


Figure 15 – Main Beam

3.4.4 Baseplate

A square welded frame made from stainless steel pipes serves as the main pedestal, providing a stable foundation for the entire turret. The stainless steel, treated with passivation and coated with paint, ensures high resistance to corrosion, making it suitable for demanding environments.

The pedestal features four motorized screw jacks located at the sides. These, combined with a self-leveling system, allow the turret to be installed on surfaces with inclinations of up to 10°. The base of the pedestal is 100 mm thick, ensuring robustness and durability.

At the rear of the platform, a Z-panel is integrated, housing inlet connectors for the SUTs' electrical cables, CAN communication, and power supply. All cables are efficiently routed through the cavities within the tubular structure, ensuring protection and organization. Additionally, an access hatch provides entry to a large compartment where the electronics are housed, allowing for easy maintenance and servicing.

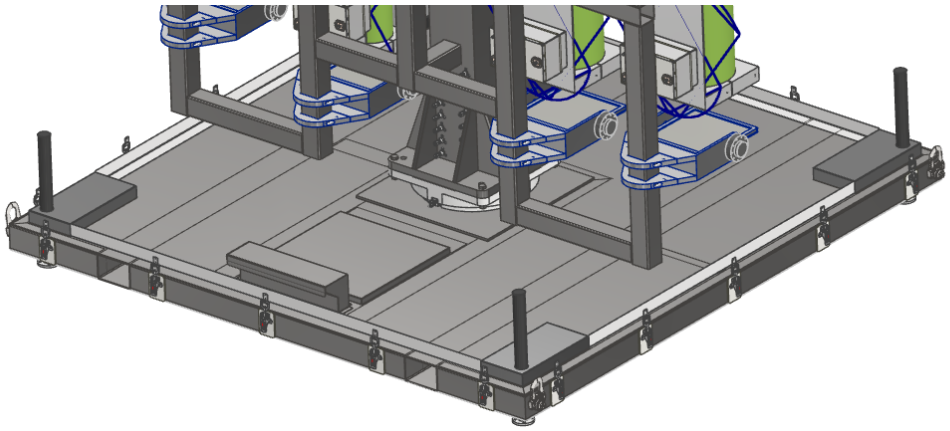


Figure 16 – Back of baseplate with Z-Panel and electronics hatch

3.4.5 The CAN Bus

The CAN interface allows broadcast communication between all connected Electronic Control Units (ECUs, also known as nodes) within the integrated electronic system through distributed control. Since the early uses of CAN in car engine management, improvements in bitrate, bandwidth, and standardization protocols (such as ISO 11898 and SAE J1939) have led to CAN utilization in various industry applications, such as factory automation, aviation, off-highway vehicles, and telematics.

The CAN technology, was selected with focused attention on the Steering Turret specific application. Following is the analysis performed to verify that CANbus is a correct solution to fully cover the ST movements performance requirements:

Assumptions (worst case, overestimated):

- 17 motors will be moved simultaneously
- All data field size (64 bits) will be used;
- 2 frames are necessary to move a single motor;

The CANbus network guarantees v_data = 1 Mbps

A CANbus datafull frame weights F_weight = 111 bits.

To move all 17 motors (8 SUTs, 8 mirrors and the Turret itself) it's needed 2 frames each, so the command weight C_weight will be:

$$CW = (FW \text{ x num of motors x num of frames}) = (111 \text{ bits x } 17 \text{ x } 2) = 3774 \text{ bits}$$

A CANbus network guarantees v_data = 1 Mbps, so the command time CT:

$$CT = (CW / v_data) = (3774 / 1\text{Mbps}) = 3{,}6 \text{ ms}$$

In order to satisfy the requirement, the movement command will give priority to the SUT movement, so:

to move the 8 SUTs the command weight CW_{SUT} will be:

$$CW_{SUT} = (F_weight \text{ x num of SUT x num of frames}) = (111 \text{ bits x } 8 \text{ x } 2) = 1776 \text{ bits}$$

the SUT command time CT_{SUT} *is*

$$CT_{SUT} = (CW_{SUT} / v_data) = (1776 / 1\text{Mbps}) = 1.7 \text{ ms}$$

The motors shift delay due to the CANbus can be estimated as:

$$SD = (\text{max_vel} * CT)$$

$$SD_{turret} = 15°/\text{s} \times 3{,}6 \text{ ms} = 0.054°$$
$$SD_{mirror} = 40°/\text{s} \times 3{,}6 \text{ ms} = 0.14°$$
$$SD_{SUT} = 30°/\text{s} \times 1.7\text{ms} = 0.051°$$

Motor	Required accuracy	Shift delay SD
Turret	0,5°	0,054°
Mirror	0,25°	0,14°
SUT	0,1°	0,051°

Table 11 – Motors performance over CANbus

The shift delay from the command start to the motor actuation is estimated to be under the accuracy required, so the reaction time of the motors can be considered negligible **[442433]**.

3.4.6 Mirrors

In order to satisfy the reflectance requirements along all the required spectra, two mirrors have been identified:

- Aluminium coated mirror: (250 - 450) ÷ (7500 – 20000) nm

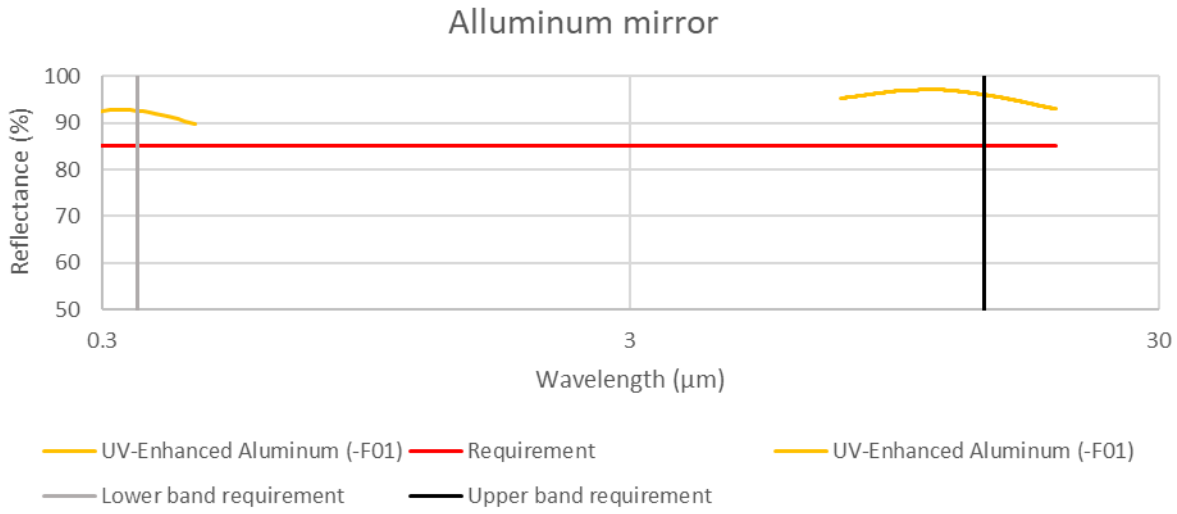


Figure 17 – Aluminium mirror performance

- Silver coated mirror: (450 - 750) nm

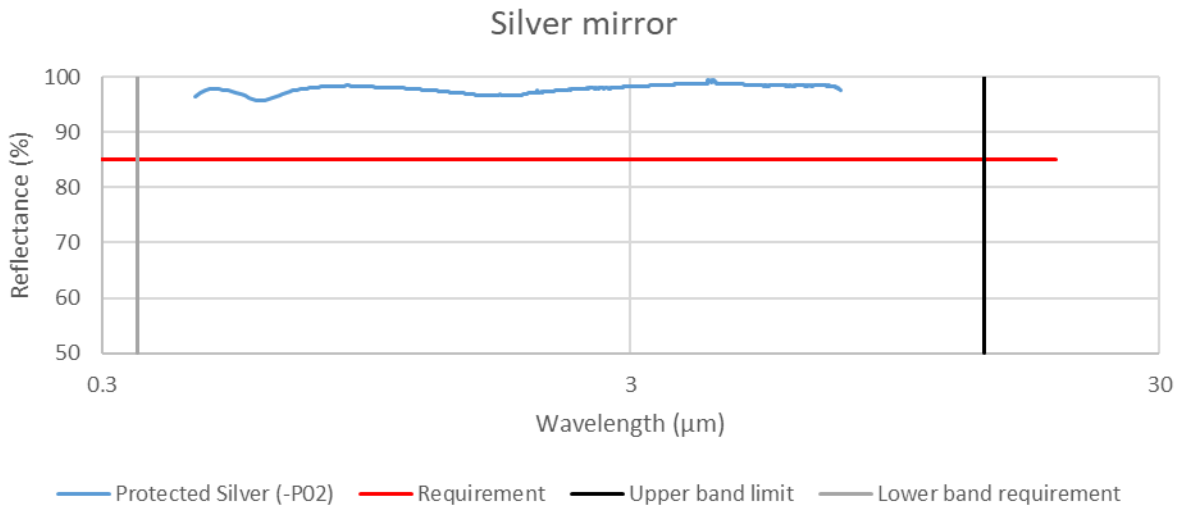


Figure 18 – Silver mirror performance

The reflectance ratio over the depicted band is over 90% satisfying the requirement [471022].

The flatness that our manufacturer can guarantee is 3λ to keep a comfortable thickness.

For the movement necessities and the shape of the box, the shape has been chosen to be a square. The evaluations give us a 26 cm side mirror to cover all the movement made by the SUT and the mirror.

The mirror is installed in a motorized frame which allow an easy installation keeping the flatness requirements:

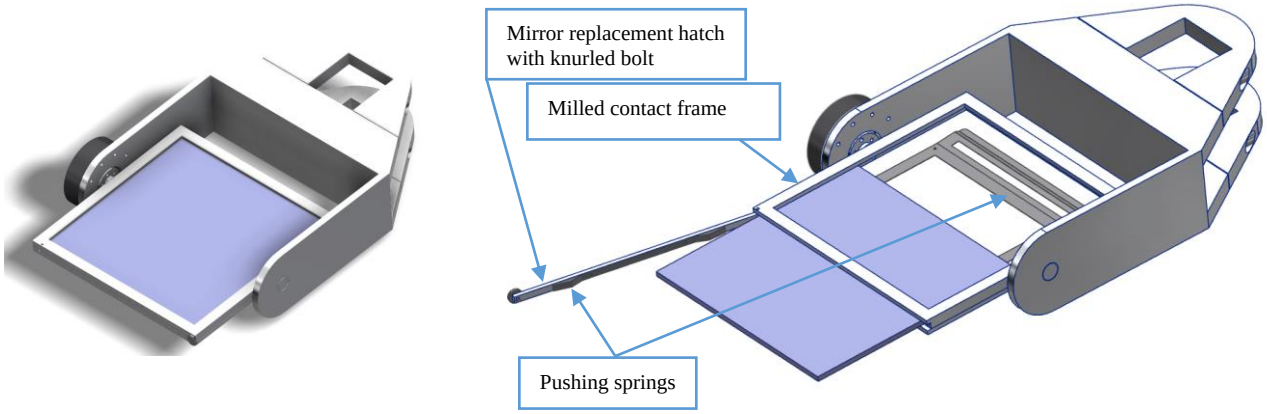


Figure 19 – Silver mirror motorized frame

The frame is done in milled aluminium, anodized and bolted to keep the flatness of the mirror and to offer the best resistance to oxidation.

3.4.7 Scheme of the control system

The system provides for the presence of motors with integrated reduction gear (HARMONIC drive with extremely low backlash), equipped with high-resolution absolute encoders. The motors include a microprocessor and an integrated drive for closing the control feedback, both in angular position and in speed.

The **master controller** manages the initialization of the motors, the configuration of the safety parameters, the synchronization and the extraction of the quantities to be monitored, sending data to the control unit. The master controller communicates with the Control Unit and provides a unified control interface for all types of motors in the system. It also manages the system diagnostics.

	Turret motor	Mirror motor	SUT motor	Camera motor
Q.ty	1 (+1)	8 (+1)	8 (+1)	1
Movement	Azimuth	Elevation	Azimuth	Elevation
Weight	Up to 1000 kg (TBC)	~ 1.0 kg	13 kg	20 kg
Range	[-140° +140°]	[-25° + 37°]	[-35° +35°]	[-5° +5°]
Accuracy	0,5°	0,25°	0,1°	0,1°
Max speed	15°/s	40°/s	30°/s	40°/s
Min speed	0,1°/s	0,1°/s	0,1°/s	0,1°/s
Max acceleration	0°/s a 15°/s in less than 1sec	0°/s a 40°/s in less than 1sec	0°/s a 30°/s in less than 5 sec	0°/s a 40°/s in less than 1sec
Inertia moment	~ 200 Kg m2	0,0052 Kg m^2	0,769 Kg m^2	0,00104 kg m^2
Torque	~ 8,3 Nm	0,00057 Nm	0,013 Nm	0,00092 Nm

Table 12 – Motors performance

3.4.8 Motors

Considering the performance requirements summarized in Table 12, following rotary actuators have been selected:

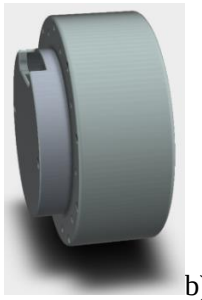
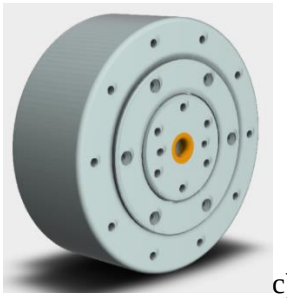
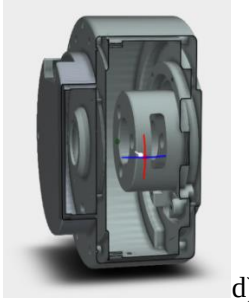
	rotary actuator
Mirror motors	Myactuator RMD-X8-20
SUT	ZeroErr eRob 70I
Camera motors	ZeroErr eRob 70I
Turret Motor	Hansheng HRT350JS-L2

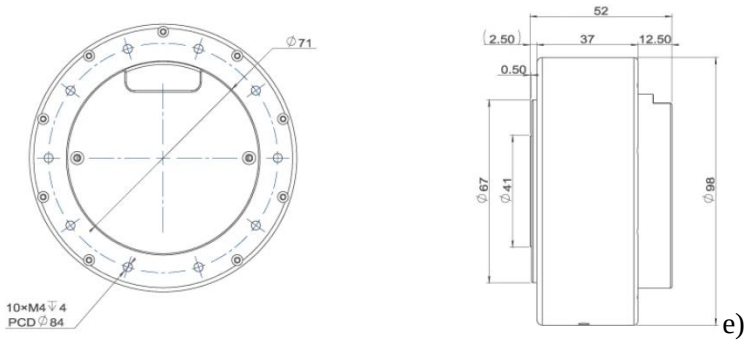
Figure 20 – Rotary actuator

3.4.8.1 Mirror motors

The driver includes various special functions, such as: power-off memory, data recording in case of power-off, bus current sampling: real-time knowledge of current-torque relationship, precise control of torque output. This actuator was selected due to its specific characteristics in terms of performance and size. The thickness makes it suitable for the mirrors movement reducing the size (width) of the Turret.



a)

b)

c)

d)



e)

Figure 21 – Mirror rotary actuator in different views

Gear Ratio		-	6:1
Input Voltage		V	48
Rated Power		W	200
Rated Torque		N.m	10
Rated Speed		RPM	190
Rated Current		A	5.2
Peak Torque		N.m	20
Peak Current		A	10.5
Efficiency		%	80
Input Encoder		-	16 bit
Output Encoder		-	14 bit
Communication Method & Baudrate		-	CAN:1M RS485:115200/500K/1M/2.5M
Axial Payload		N	985
Radial Payload		N	1250
Inertia		$Kg.cm^2$	20
Pole Pair		-	20
Back Drive Torque		N.m	0.4
Blacklash		arc min	10
Weight	N (without brake)	g	780
	B (with brake)	-	/

*** Rated torque test method:** When the ambient temperature is 24 degrees Celsius (no other heat dissipation methods), the test is performed at the rated speed. The motor torque reaches temperature balance under the condition of a temperature rise of 60 degrees Celsius, and the long-term working point is the rated torque value of the motor.

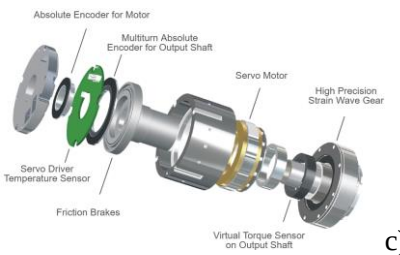
Table 13 – RMD-X8-20 product parameter

3.4.8.2 SUT and Camera motor


a)



b)



c)

Figure 22 – SUT rotary actuator in different views

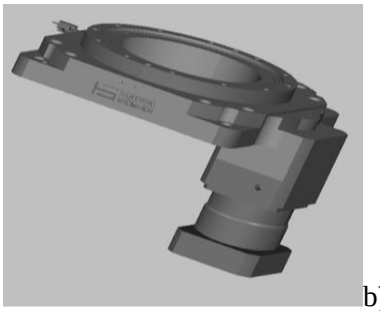
Hollow Bore Diameter	18mm	
Precision	Repeatability/Absolute Accuracy: ±7/±15 arc sec or ±10/±25 arc sec*	
Electrical Interface	Communication Protocol: EtherCAT/CANopen/Modbus Output-side Encoder Resolution:19/20Bit	
Strain wave gear-ratio	14-50	
Peak torque for start and stop (Nm)	23	
Permissible max.value at average load torque(Nm)	9	
Rated torque(Nm)	7	
Permissible maximum momentary torque (Nm)	46	
Max. output rotational speed(RPM)	60	
Motor power (W)	100	
Power Input Voltage (V)	48V (±10%)	
SWG Output Side Inertia (g-mm2)	68484	
SWG Output Side Mass (kg)	0.166	
Without Brake	SWG Input Side Inertia (g·mm²)	22674
	SWG Input Side Mass (kg)	0.087
	OD * Length (mm)	Φ70x71
	Total Mass (kg)	0.82
With Brake	SWG Input Side Inertia (g·mm²)	26128
	SWG Input Side Mass (kg)	0.1
	OD * Length (mm)	Φ70x71
	Total Mass (kg)	0.88
Common parameters	ID:18mm, IP grating：IP54, Brake model:friction brake	
Optional device	Communication type：EtherCAT/CANopen/Modbus; Output encoder resolution：19Bit Repeatability/Accuracy：±7/±15 arc seconds or ±10/±25 arc seconds.	

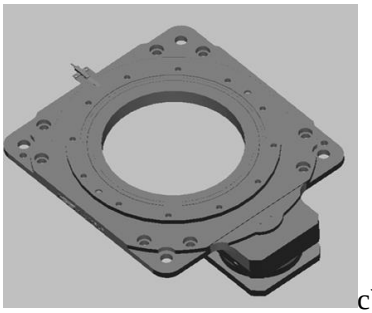
Table 14 – eRob 70I product parameter

3.4.8.3 Turret Motor

Following Figure 23 show the Turret motor layout. This rotary actuator is suitable to move the Turet with the required precision, also considering the Turret full weight up to 1000 kg. Selection was driven by the size of the central hole, necessary for the cable routing.


a)


b)


c)

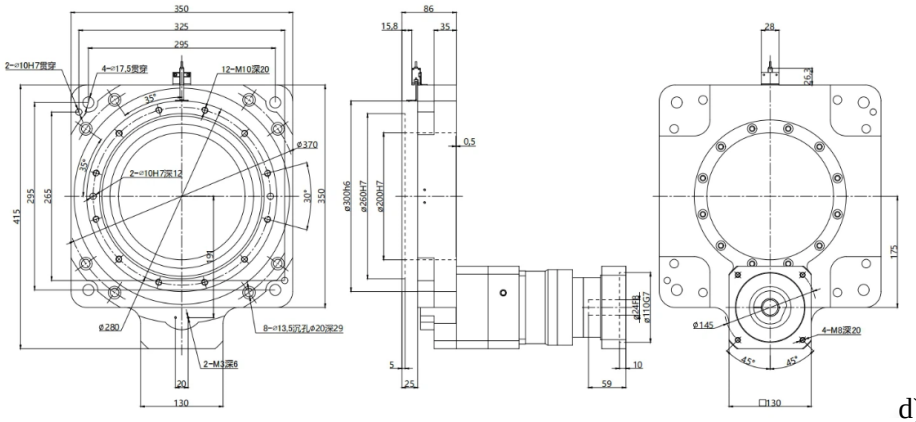


Figure 23 – Rotary actuator in different views

Gear Ratio	200 300 400 500 700 1000	Allowable Moment of inertia	420 N.m	Protection Level	IP50
Rated Output Torque	890 N.m	Backlash	None-backlash (adjustable)	Photoelectric Switch	Standard Configure
Max Output Torque	3 Times Rated Output Torque	Torsional Rigidity	105 N.m/arcmin	Lubrication	Long-term Lubrication
Bearing Structure	Cross Roller Bearing	Precision Lifespan	30000 h (intermittent)	Transmission Efficiency	87%
Allowable Axial Load	17500 N	Positioning Accuracy	≤4 arcmin	Noise	≤67dB
Allowable Radial Load		Matched Servo Motor	1KW-3KW	Installation Method	Any
Allowable Disc Speed	200 rpm (intermittent)	Matched Stepper Motor	110-130 Stepper	Designed Loading	450kg
Concentricity	≤0.015 mm	Repeat Positioning Accuracy	≤60 arcsec	Weight	
Parallelism	≤0.035 mm				

Table 15 HRT350JS-L2 product parameter

3.4.9 Cabling and connectors

The Turret will provide a single panel for all external interface to the rear of the turret platform: the Z-Panel.

Following connectors will be available on the z-panel:

- MIL-DTL-D38999/24_G75PN connector, for power supply;
- MIL-DTL- D38999/24_D35SB connector for the communication between the Turret and the CU. The connector provides 37 pin, used as follow:
  HDFDI: 2 wires, for the Aiming camera data link
  RS-485: 3 wires (TBC. Up to 9 wires could be requested) for the Aiming camera control link
  Discrete: 6 wires considering a common ground
  Ethernet 8 wires (for the link between the CU switch and the Platform switch).
  Ethernet 8 wires for the Giga Ethernet as common link for IR camera video data;
- 8 MIL-C-26482 standard Souriau Series 851 36G 20-41P 50 connector, for SUT data and power.

For details on electrical and data interface, see [R.05].

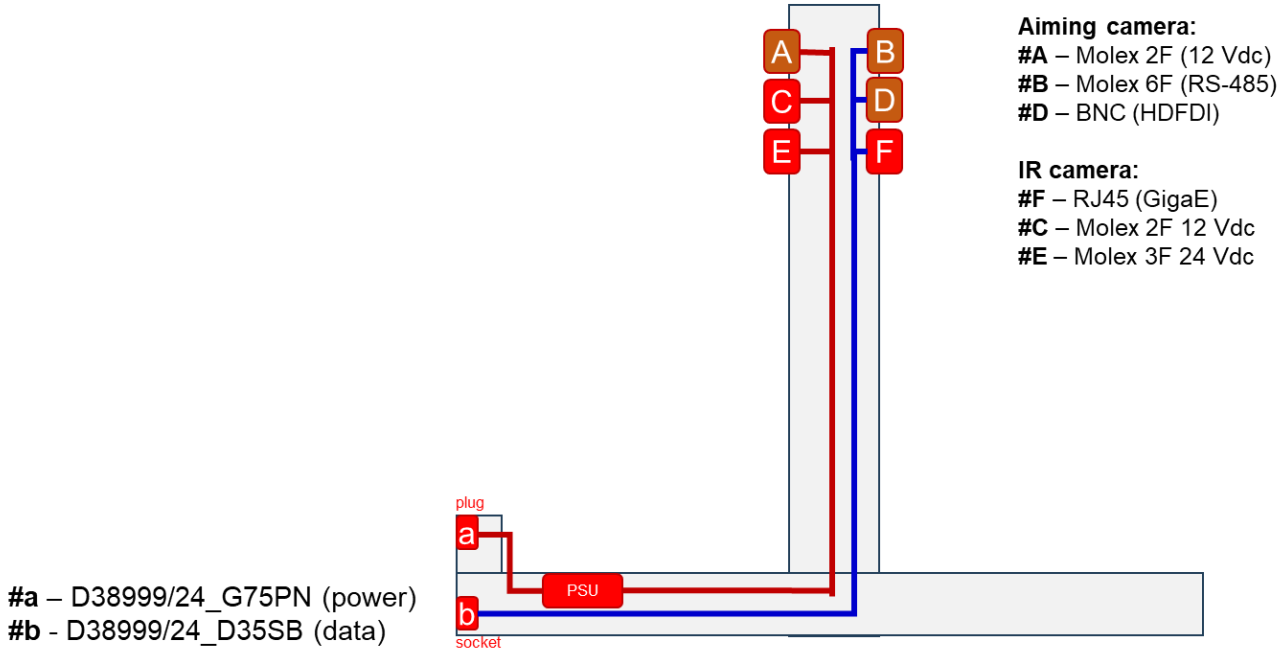


Figure 24 – Camera power/data routing

3.4.9.1 Aiming Camera Connections

The aiming camera, requires following power/data interfaces:

Following connectors will be available on top of the Turret, close to the camera block:

Type	Model	Interface		Connector on board	Connector at the top of the Turret
Aiming camera		Command			Molex 6 pin female (TBC)
		Video			HD-SDI BNC female (TBC)
		Power supply	**12 Vdc**		Molex 2 wire female (TBC)

Table 16 – Aiming camera power, control and video connectors

The use of Molex connectors is discouraged for devices working outdoors. Although the connector panel at the top of the turret will be protected from rain, a different solution should be considered.

3.4.9.2 IR Camera Connections

Different link types are requested for the IR cameras (see Table 17).

Type	Model	Interface		Connector on board (camera)	Mating connector (on turret)
FLIR	SC7000[1]	GigaEthernet		RJFTV22N03100BTX	RJFTV6N
		Camera LINK		Amphenol TVPS00FCI1726P	
		Video		LEMO ERP1S306CLN	
		Power supply	**12 Vdc 5 A**	LEMO ECP1S302CLN	LEMO FFA1S302CLAC52

[1]Industrial module has been considered.					
FLIR	x6580sc 66101-0101	GigaEthernet		RJ45 female	RJ45 male
		USB			
		FireWire			
		Power supply[2]	**24 Vdc**	Unspecified	TBD
[2]The power supply is provided with the camera					
InfraTec	ImageIR 9400	10 GigaEthernet		RJ45 female (TBC)	RJ45 male (TBC)
		HDMI[3]			
		CAMLink[3]			
		GigE[3]			
		Power supply	**24 Vdc**	Unspecified	TBD
[3]Depending on model and specific configurations.					
InfraTec	ImageIR 8800	GigE, 10 GigE		RJ45 female (TBC)	RJ45 male (TBC)
		10 GigE[4]			
		2 × CAMLink[4]			
		HDMI[4]			
		Power supply	**24 Vdc**	Unspecified	TBD
[4]Depending on model					

Table 17 – Additional IR camera power and data link

Both 12Vdc and 24 Vdc connector will be available on top of the Turret, in order to match a wide range of additional cameras. A RJ45 connector for the GigaE video data will be available.

Type	Model	Interface		Connector on board	Connector at the top of the Turret
Additional (IR) camera		Giga Ethernet		Giga Ethernet	RJ45
		Power supply	**24 Vdc**	Undefined. It could vary according the camera model so an adapter could be necessary	Molex 3 pin female
		Power supply	**12 Vdc**	according the camera model so an adapter could be necessary	Molex 2 pin female

Table 18 – Additional IR camera power and video connectors

3.4.9.3 SUT Connections

Following split cables are currently used to connect the SUTs (all types) to the CBOX.

Split cable name	SUT split cables			
	Connector to CBOX	SUT connector	Spin Module	BNC
BIT-LC	Amphenol 62GB 12E20-41SN	Amphenol 62GB 12E18-32SN	Amphenol 62GB-12E12-10PN	
FLENS	Amphenol 62GB 12E20-41SN	Amphenol 62GB 12E16-26SN	Amphenol 62GB-12E12-10PN	

Augustiner	Amphenol 62GB 12E20-41SN	62GB 20-41PN	62GB-12-10PN	
		SUT (X22)	**SUT (X23)**	
BECKS	Amphenol 62GB 12E20-41SN	unspecified	unspecified	
		LC SEEKER	**Spin Module**	
JEVER	Unspecified* 41-WAY umbilical connection	Unspecified 19 WAY MIL F	9-WAY 'D' PLG	
		PAULINE SEEKER	**Spin Module**	
Schneider	Unspecified* 41-WAY umbilical connection	Unspecified 32 WAY plug	Unspecified 32 WAY socket	(Rejection and Bias zones)
		WEASEL Seeker (P1003)	**Telemetry control (P901)**	
Bravo HV	Unspecified* 41-WAY umbilical connection	Unspecified 34 WAY	15 -WAY 'D'	
***It is supposed to be an Amphenol 62GB 12E20-41SN (or equivalent)				

Table 19 – Turret connector mating SUTs connectors

A common MIL-C-26482 **41 way** "CBOX" connector is provided on the Turret side for all SUTs, where the specific SUT split-cable can be attached on.

Interface	Connector on platform base	Connector on Turret side
SUT control and power	Souriau Series 851 07EC 20-41P 50	Souriau Series 851 07EC 20-41S 50

Following Figure 25 shows the SUTs cable routing inside the turret and the relevant interface (in red); in blue colour the standard split cables in use for different SUT type. In the schema it is supposed the CBOX interface is common for all SUT types, so it is replicated on the turret side, close to the SUT position, by means of a **Souriau Series 851 07EC 20-41P 50.**

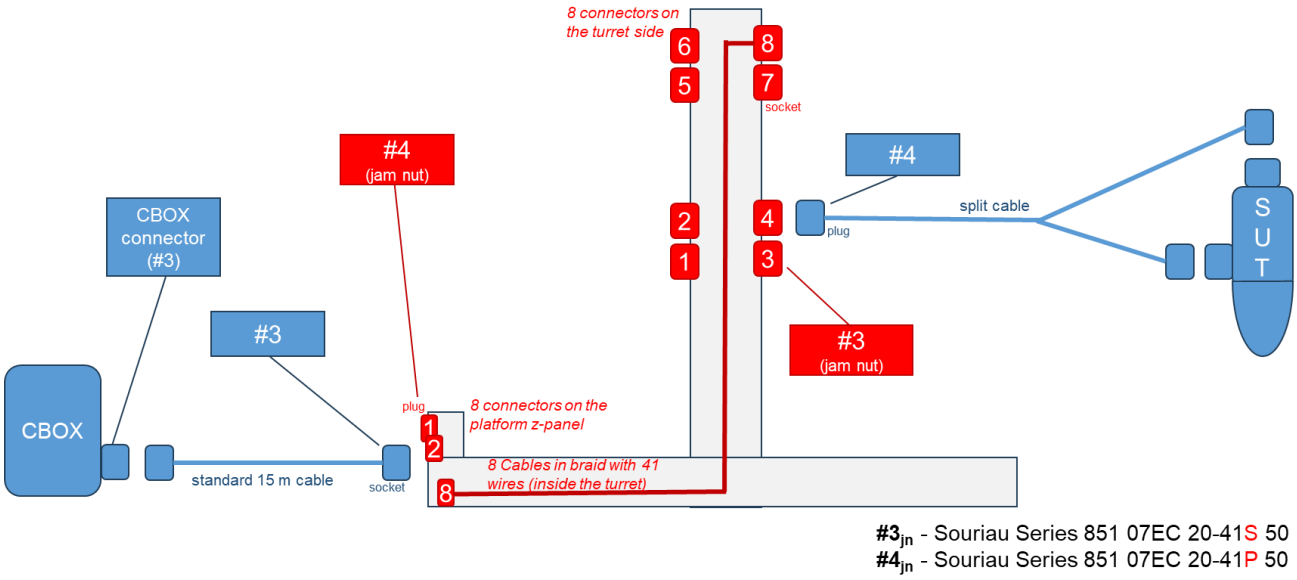


Figure 25 – SUTs cable extension routing

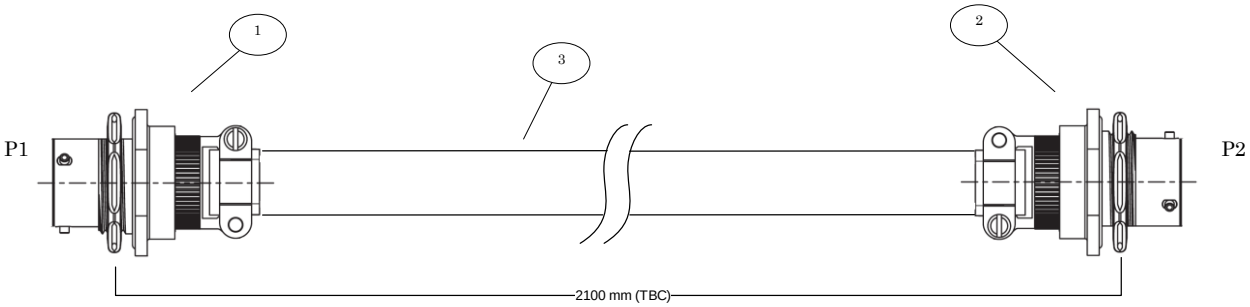


Figure 26 – SUTs cable extension layout

3.4.10 The Nitrogen Distribution

The Turret provides 16 Nitrogen input connectors at the Turret base and 16 output connectors in the upper part. Figure 27 shows 8 connectors placed at one side of the main beam while the others are located at the other side of the beam, close to the SUTs.

There are two nitrogen distribution lines for each SUT.

Figure 27 b) and c) also shows how the nitrogen connectors are distributed along the main beam.

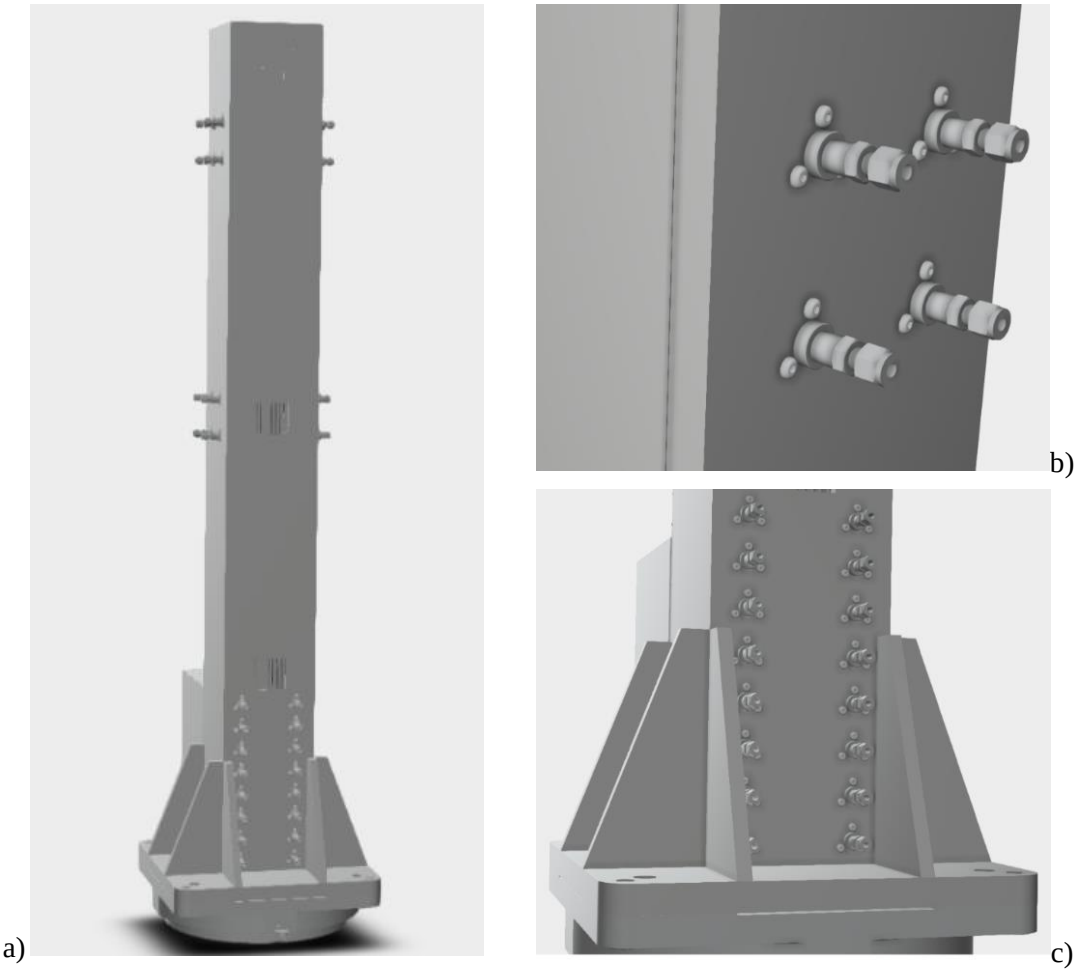
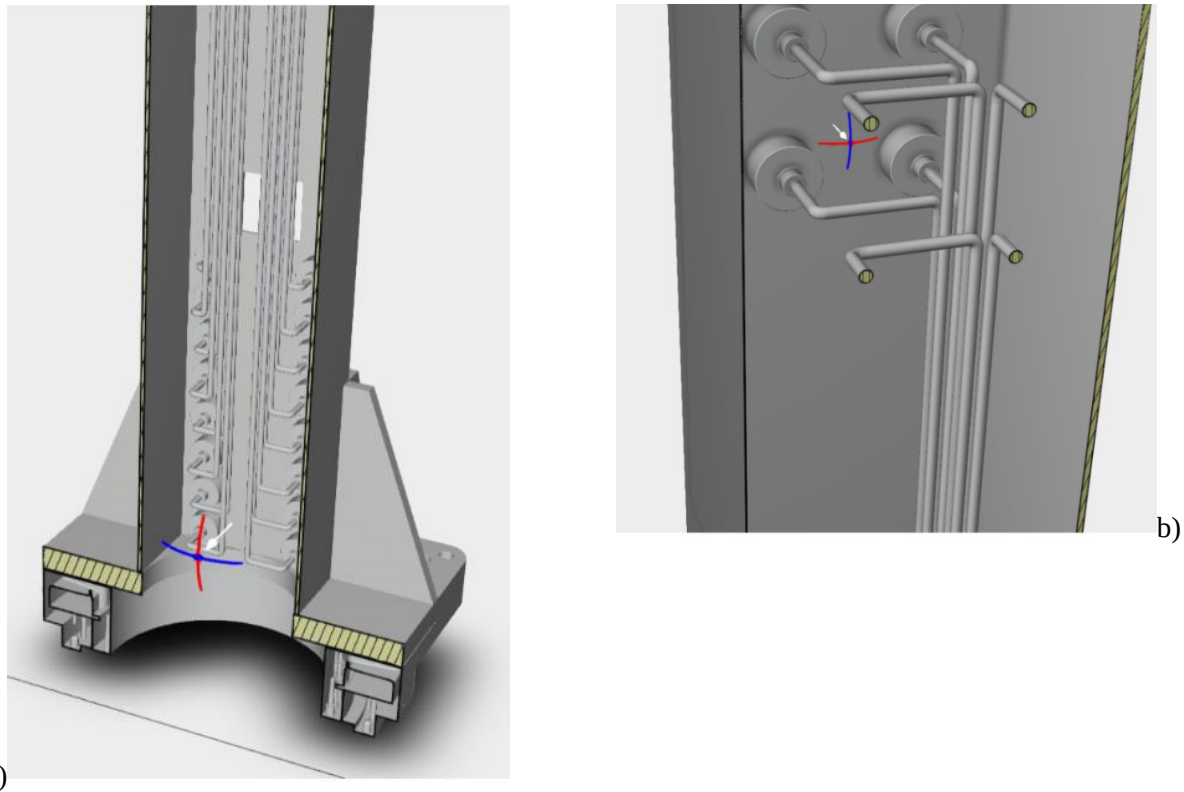


Figure 27 – Nitrogen distribution. External (a) A detail of the distribution of nitrogen connectors in the upper (b) and lower (c) part of the turret is shown.

In order to make the product more durable and reliable, stainless-steel pipes have been selected for the construction of the nitrogen distribution system. Stainless-steel pipes pass inside the main beam so that there is no need of scheduled maintenance. Pipes routing inside the main beam of the Turret is shown in Figure 28.



a) b)

Figure 28 - Detail of pipes routing at the base of the central pole of the Turret (a) and in its upper part (b).

Considering that the lines used for nitrogen distribution up to the SUTs are made with a flexible MCS 6400 tube (black lines), the connectors selected to ensure their connection to the pipes inside the turret will be compatible with this product line. Figure 29 shows the mechanical features of the selected blackline connectors.

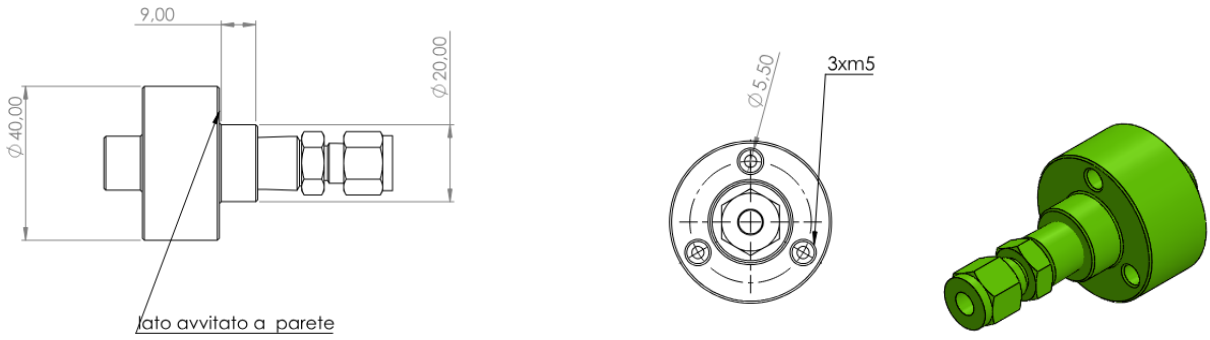


Figure 29 –Blackline connector selected for the nitrogen distribution system.

These connectors will be screwed to the wall of the main beam as shown previously in Figure 27.

3.4.11 Self-levelling system

The platform levelling system has been considered as a "low dynamic" system, that is, it reaches the "zero level" (that is, all the error angles are zeroed) in a short but not very short time: this is to avoid subjecting the payload

that will be placed on the platform to excessive lateral accelerations (and related damped oscillation). The following specifications will apply:

- The leveling system proceeds from a maximum overall offset of 10°.
- The maximum achievable precision is 0.004° (TBC)
- The leveling system is of the "Top Leg" type

In the block diagram shown in figure, at the top you can see the reference system from which the zeroing system derives. Then there is a symbolic "flow chart" that shows the logical flow of data until freezing.

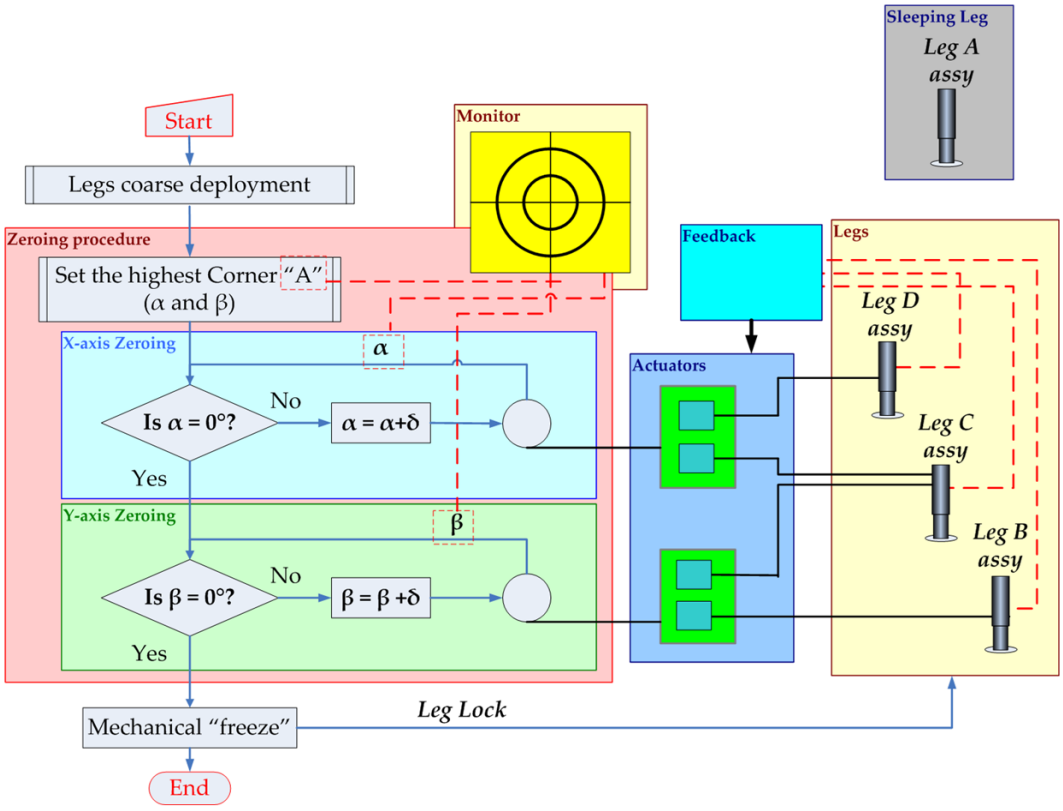


Figure 30 – Levelling Functional Block Diagram

Top Leg procedure:

1. First the highest leg is identified (Leg A in the image)
2. Then the error about the local X-axis (alpha angle) is reset to zero.
3. The error about the local Y-axis (beta angle) is then reset to zero.
4. Once the error is reset, the leg movement is "frozen" and blocked.

In the method we propose, the so-called "top leg" system consists of identifying the leg that is highest with respect to the other three: the others (two by two) are then activated until they reach the same height as the aforementioned highest leg. Note that the higher leg will not move during the entire procedure, acting as a pivot.

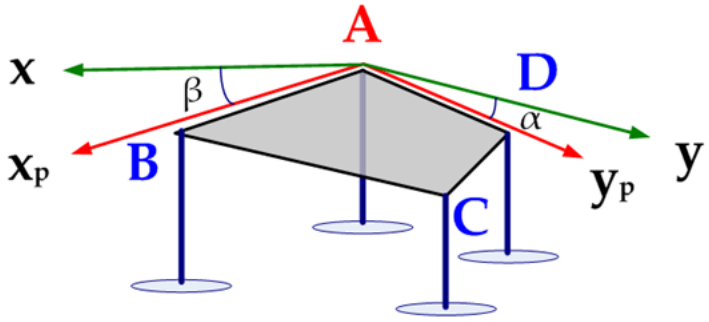


Figure 31 – Levelling Angles and Legs identification

The control system features a "synthetic bubble" that graphically shows the operator how the platform is approaching the zero angular error point. SEIKA SB2i-NB3 XY inclinometers were chosen, which allow for high precision as a function of the maximum measurable angle.



Figure 32 – Levelling inclinometers package

3.4.12 Laser pointers

The calibration process involves the usage of laser pointers to properly align the system. For this purpose, the Turret will be equipped with pointer holders that guarantee the parallelism between the LoS of the SUT and the laser pointing vector.

Each laser is headed towards the mirror, thus its dimensions were selected considering also the presence of the laser which is integrated in the SUT mounting bracket, as shown in Figure 33.

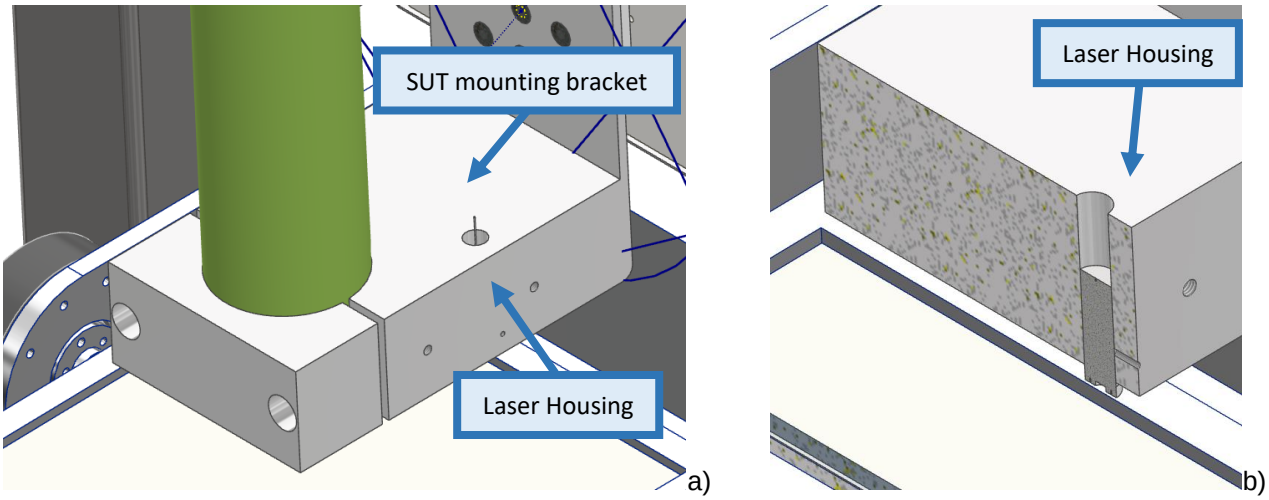


Figure 33 - SUT mounting bracket with integrated a laser for calibration (a). A section of it is shown in (b).

The alignment laser module that will be installed in the Turret have a wavelength of 635 nm, an output power of 5 mW and are powered via a 5 VDC source (Figure 34).

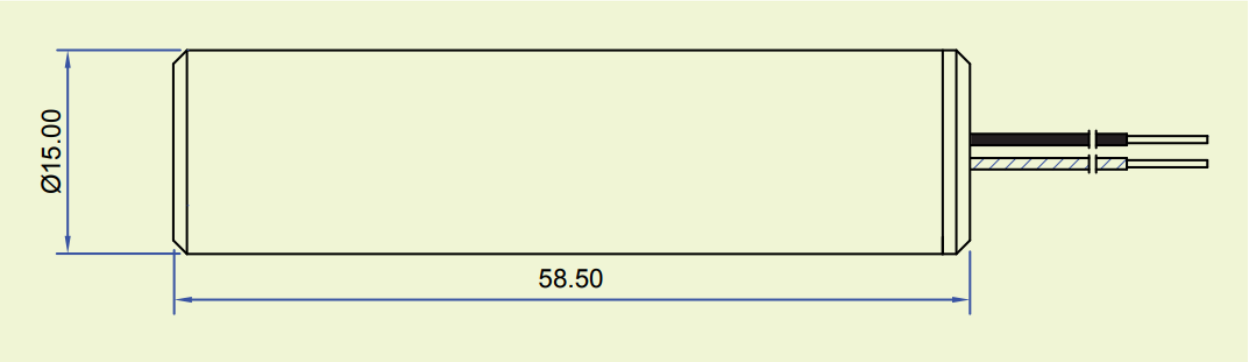


Figure 34 - The alignment laser diodes module selected for the project is depicted in (a). Mechanical features of the component are shown in (b).

Considering its purpose, the laser has been chosen keeping the beam divergence as low as possible. The variation of the beam diameter with respect to distance is shown in Figure 35. These measurements were performed at 25°, so at different temperatures could be different. The declared operating case temperature goes from -10°C to +45°C.

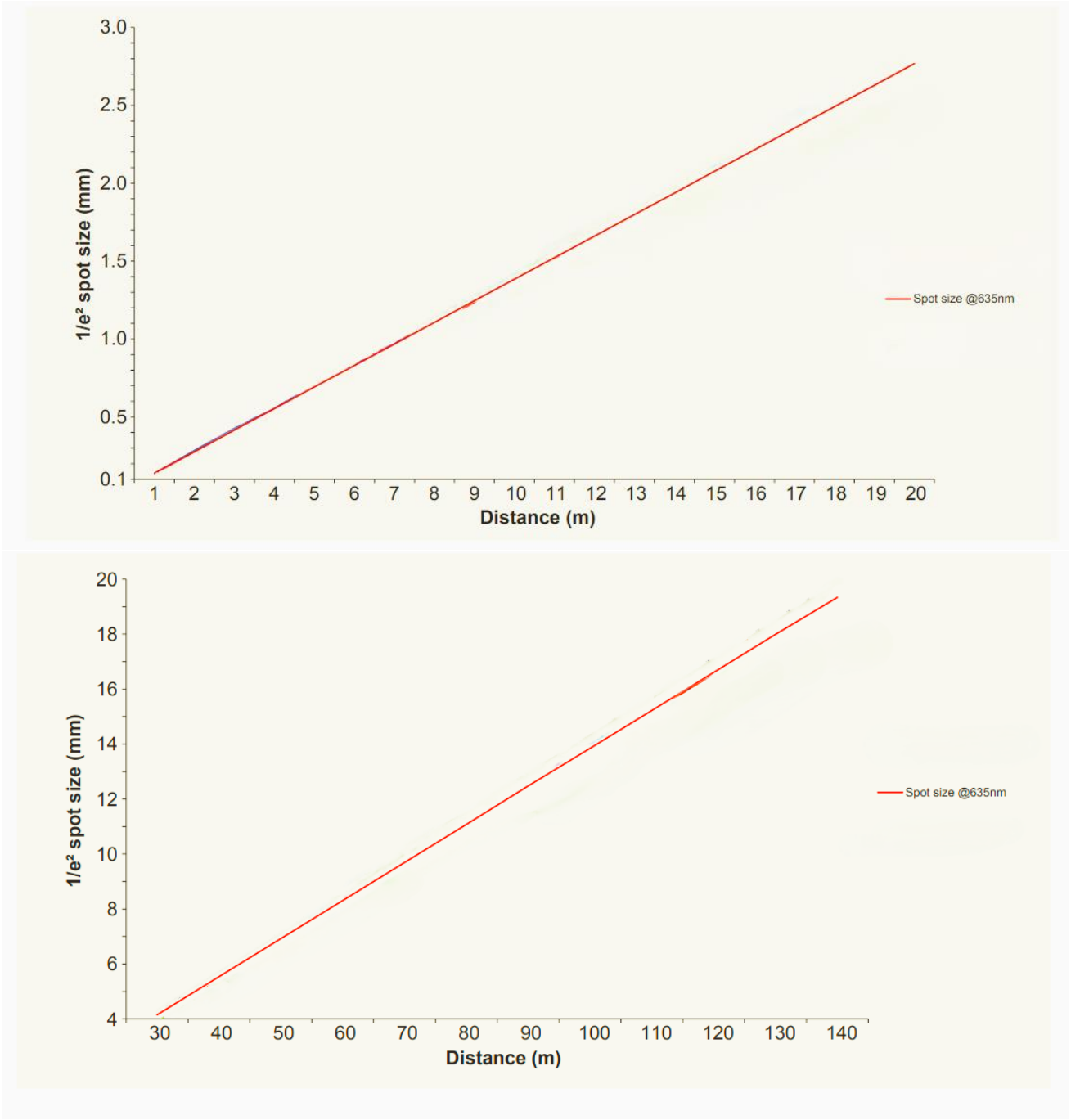


Figure 35 – Variation of beam diameter with respect to distance.

The calibration panel where the laser spot will be projected is not supplied.

4 TRACEABILITY MATRIX

N/A to the present issue of the document.

A.1 APPENDIX A

A.1.1 Built-in test

Built-In Test	CU	Turret	PBIT	CBIT	IBIT
Health status of the Aiming Cameras	X		X	X	
Health status of the CHARM-100NX tracking board	X		X	X	
Health status of the Power Distribution Unit	X		X	X	
Check Input voltage to the Turret	X		X	X	
CU-Turret Interface	X	X	X	X	
Health status of the Turret controller program		X	X	X	
Turret Temperature status		X	X	X	
Health Status of the motors		X	X	X	
Test Angular limits		X		X	X
Health status of the CU Computer (storage, OS, DIO Board, …)	X		X	X	X
Health status of the CU controller program	X		X	X	

Table 20 – Built-In Tests

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ALMA Sistemi srl
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